

The pyramid — height, apothem and surface area

Three lengths, one right triangle, and every pyramid question follows.

LESSON 33–34

2 × 40 MIN

GEOMETRY 11, §15

IGCSE E13.X

LESSON CLOCK

14'

24'

24'

20'

16'

4

The elements	14 min
The two right triangles	24 min
Surface area	24 min
Working backwards	20 min
Practice	16 min
Homework	4 min

LEARNING OBJECTIVES

- Name the elements of a pyramid and of a regular pyramid.
- Relate the height, apothem, slant height and lateral edge.
- Compute the lateral and total surface area.
- Solve for a missing element from a given area.

TEXTBOOK

National	pp. 161–172
Cambridge	Core & Extended, pp. 323–330
Workbook	Exercise 15.1

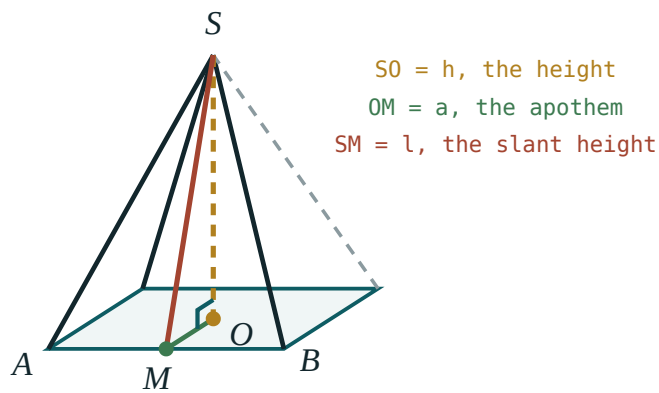
01

Explanation

The elements

REGULAR PYRAMID

Its base is a **regular polygon** and its apex lies directly above the base **centre**. Then all the lateral edges are equal, and all the lateral faces are congruent isosceles triangles.



$SO = h$, the height
 $OM = a$, the apothem
 $SM = l$, the slant height

$l^2 = h^2 + a^2 \cdot S_{lat} = \frac{1}{2}Pl$

The height to the centre, the apothem across the base,
the slant height up the face.

Why the lateral edges are equal: their projections are all radii of the circumcircle of the base, hence equal — and equal projections give equal obliques, by Grade 10 Quarter III.

The two right triangles

Every pyramid calculation lives in one of two right triangles, both with the height h as a leg:

$l^2 = h^2 + a^2$ (slant height, apothem) $e^2 = h^2 + R^2$ (lateral edge, circumradius)

SQUARE BASE, SIDE s	VALUE
apothem a	$\frac{s}{2}$
circumradius R	$\frac{s\sqrt{2}}{2}$
slant height l	$\sqrt{h^2 + (s/2)^2}$
lateral edge e	$\sqrt{h^2 + s^2/2}$

THE APOTHEM IS NOT THE CIRCUMRADIUS

For a square of side 10: apothem 5, circumradius $5\sqrt{2} \approx 7.07$. Using the wrong one is the standard error, and it makes the slant height and the lateral edge swap places.

Surface area

$$S_{lat} = \frac{1}{2} P \cdot l \quad S_{total} = B + \frac{1}{2} Pl$$

Each lateral face is a triangle of base one side and height l ; there are as many as the base has sides, so the total is $\frac{1}{2}$ of the perimeter times l .

Example. Square base 10, height 12: $a = 5$, $l = 13$, $P = 40$, $B = 100$. So $S_{lat} = 260$ and $S = 360$.

ONLY FOR A REGULAR PYRAMID

$\frac{1}{2} Pl$ assumes every lateral face has the same slant height. For an irregular pyramid the faces must be computed one at a time.

Working backwards

Given the area, recover a length by reversing the formula:

$$l = \frac{2S_{lat}}{P} \quad h = \sqrt{l^2 - a^2}$$

Example. A square pyramid with base 12 has lateral area 240. Then $l = \frac{480}{48} = 10$,

$a = 6$, so $h = \sqrt{100 - 36} = 8$.

02 Worked examples

EXAMPLE 1 A regular square pyramid has base edge 10 and height 12. Find the slant height, the lateral edge and the total surface area.

$$① \quad a = 5, \text{ so } l = \sqrt{144 + 25} = 13.$$

$$② \quad R = 5\sqrt{2}, \text{ so } e = \sqrt{144 + 50} = \sqrt{194} \approx 13.9.$$

$$③ \quad S_{lat} = \frac{1}{2} \times 40 \times 13 = 260$$

$$④ \quad S = 100 + 260 = 360$$

ANSWER

$$l = 13, e \approx 13.9, S = 360$$

EXAMPLE 2 A regular hexagonal pyramid has base edge 6 and slant height 10. Find the lateral surface area and the height.

$$① \quad P = 36, \text{ so } S_{lat} = \frac{1}{2} \times 36 \times 10 = 180.$$

$$② \quad \text{Apothem of a regular hexagon of side 6: } 3\sqrt{3} \approx 5.196.$$

$$③ \quad h = \sqrt{100 - 27} = \sqrt{73} \approx 8.54$$

ANSWER

$$S_{lat} = 180; h \approx 8.54$$

EXAMPLE 3 A square pyramid with base 12 has lateral surface area 240. Find its height.

$$① \quad l = \frac{2 \times 240}{48} = 10$$

$$② \quad a = 6$$

$$③ \quad h = \sqrt{100 - 36} = 8$$

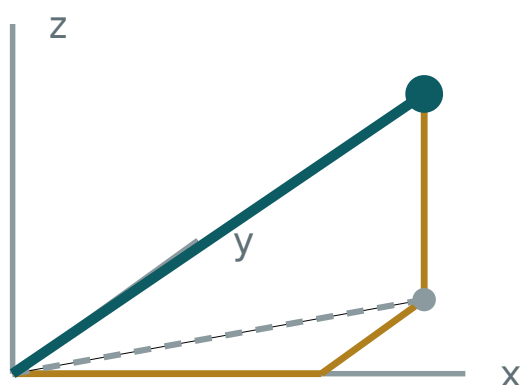
ANSWER

8

03 Interactive model

Build the pyramid net from Quarter I and measure the slant height directly.

Inside a pyramid



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Pyramid	Piramida	Пирамида
Apex	Uchi	Вершина
Regular pyramid	Muntazam piramida	Правильная пирамида
Height of a pyramid	Piramida balandligi	Высота пирамиды
Apothem of the base	Asos apofemasi	Апофема основания
Slant height	Yon apofema	Апофема
Lateral edge	Yon qirra	Боковое ребро
Lateral surface area	Yon sirt yuzasi	Площадь боковой поверхности
Base perimeter	Asos perimetri	Периметр основания

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

In a regular pyramid the apex is above:

l^2 equals:

S_{lat} of a regular pyramid:

For a square base of side 10 the apothem is:

And the circumradius is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Square base 10, height 12: the slant height

ANSWER 13

2. Square base 8, height 3: the slant height

ANSWER 5

3. Square base 6, apothem

ANSWER 3

4. Square base 6, circumradius

ANSWER $3\sqrt{2}$

5. S_{lat} with $P = 24$, $l = 10$

ANSWER 120

6. S_{total} with $B = 36$, $P = 24$, $l = 10$

ANSWER 156

7. Square base 10, height 12: the lateral area

ANSWER 260

Medium · the standard question

7 PROBLEMS

1. Square base 10, height 12: total area

ANSWER 360

2. Square base 10, height 12: the lateral edge

ANSWER $\sqrt{194} \approx 13.9$

3. Hexagonal base 6, slant height 10: lateral area

ANSWER 180

4. Same: the height

ANSWER $\sqrt{73} \approx 8.54$

5. Square base 12, lateral area 240: the height

ANSWER 8

6. Square base 16, height 6: total area

ANSWER 576

7. Square base 14, lateral edge 25: the height

ANSWER 24

Hard · stretch and reason

7 PROBLEMS

1. A regular triangular pyramid, base edge 6, height 4: the total area

ANSWER $9\sqrt{3} + 3\sqrt{57} \approx 38.2$

2. Square base s , height h : prove $e^2 = h^2 + \frac{s^2}{2}$

ANSWER $R = \frac{s\sqrt{2}}{2}$

3. A square pyramid of total area 384 with base 12: the height

ANSWER 8

4. A regular hexagonal pyramid, edge 4, height 6: the volume

ANSWER $48\sqrt{3} \approx 83.1$

5. A pyramid's lateral faces make 60° with the base, base edge 10: the height

ANSWER $5\sqrt{3}$

6. Show that $e > l > h$ in every regular pyramid

ANSWER Each is the hypotenuse of the previous

7. A square pyramid with all edges equal to a : its height

ANSWER $\frac{a}{\sqrt{2}} \approx 0.707a$

07 Homework

Homework — 5 tasks

Say which triangle you are using — apothem or circumradius — on every line.

1. A regular square pyramid has base edge 16 and height 15. Find the slant height, the lateral edge and the total surface area.
2. A regular hexagonal pyramid has base edge 5 and slant height 12. Find the lateral area and the height.

3. A square pyramid with base 20 has lateral area 520 . Find the slant height and the height.
4. A regular triangular pyramid has base edge 8 and height 6 . Find the slant height and the total surface area.
5. Prove that in a regular pyramid the lateral edges are equal.

The frustum of a pyramid

A pyramid with its top cut off — and the trapezium faces that result.

LESSON 35

1 × 40 MIN

GEOMETRY 11, §16

EXTENSION

LESSON CLOCK

8'

14'

14'

6'

3'

What a frustum is	8 min
Surface area	14 min
Volume	14 min
Recovering the pyramid	6 min
Homework	3 min

LEARNING OBJECTIVES

- Describe the elements of a frustum of a pyramid.
- Compute its lateral and total surface area.
- Compute its volume.
- Recover the original pyramid from the frustum.

TEXTBOOK

National	pp. 173–178
Cambridge	Core & Extended, pp. 331–334
Workbook	Exercise 16.1

01

Explanation

What a frustum is

Cut a pyramid with a plane parallel to its base and remove the small pyramid on top. What is left is a **frustum**. Its two bases are **similar** polygons in parallel planes, and its lateral faces are trapezia.

THE KEY RATIO

If the cut is at height k of the way up, the small pyramid is similar to the whole with scale factor k — so its base edges are k times, its area k^2 times, and its volume k^3 times.

Surface area

$$S_{lat} = \frac{1}{2}(P_1 + P_2) \cdot l$$

where P_1, P_2 are the two base perimeters and l is the slant height of the frustum — the height of one trapezium face. This is the trapezium area formula, once per face, collected.

$$S_{total} = B_1 + B_2 + \frac{1}{2}(P_1 + P_2)l$$

Example. A frustum of a square pyramid with bases 10 and 4, slant height 5: $S_{lat} =$

$$\frac{1}{2}(40 + 16)(5) = 140; S = 100 + 16 + 140 = 256.$$

Volume

$$V = \frac{h}{3}(B_1 + B_2 + \sqrt{B_1 B_2})$$

It is the big pyramid minus the small one, simplified. The middle term $\sqrt{B_1 B_2}$ is the geometric mean of the two bases, and it is what makes the formula more than an average.

Example. Bases 10 and 4 (so $B_1 = 100$, $B_2 = 16$), height 4:

$$V = \frac{4}{3}(100 + 16 + 40) = \frac{4}{3}(156) = 208$$

NOT THE AVERAGE OF THE TWO BASES

$\frac{h}{2}(B_1 + B_2)$ would give 232 — too big. The frustum tapers, and the geometric mean is what accounts for it.

Recovering the pyramid

If the bases are a and b with $b < a$, the scale factor is $k = \frac{b}{a}$. The full pyramid's height H satisfies

$$H - h = kH \Rightarrow H = \frac{h}{1 - k}$$

For bases 10 and 4 with $h = 4$: $k = 0.4$, so $H = \frac{4}{0.6} \approx 6.67$. Check the volume: $\frac{1}{3}$

$$(100)(6.67) - \frac{1}{3}(16)(2.67) = 222 - 14.2 \approx 208 \checkmark.$$

02 Worked examples

EXAMPLE 1 A frustum of a square pyramid has bases 10 and 4 and slant height 5 . Find its total surface area.

$$(1) \quad P_1 = 40, P_2 = 16$$

$$(2) \quad S_{lat} = \frac{1}{2}(56)(5) = 140$$

$$(3) \quad B_1 = 100, B_2 = 16$$

$$(4) \quad S = 256$$

ANSWER

256

EXAMPLE 2 The same frustum has height 4 . Find its volume.

$$(1) \quad \sqrt{B_1 B_2} = \sqrt{1600} = 40$$

$$(2) \quad V = \frac{4}{3}(100 + 16 + 40)$$

$$(3) \quad = \frac{4}{3} \times 156 = 208$$

ANSWER

208

EXAMPLE 3 Find the height of the original pyramid.

$$① \quad k = \frac{4}{10} = 0.4$$

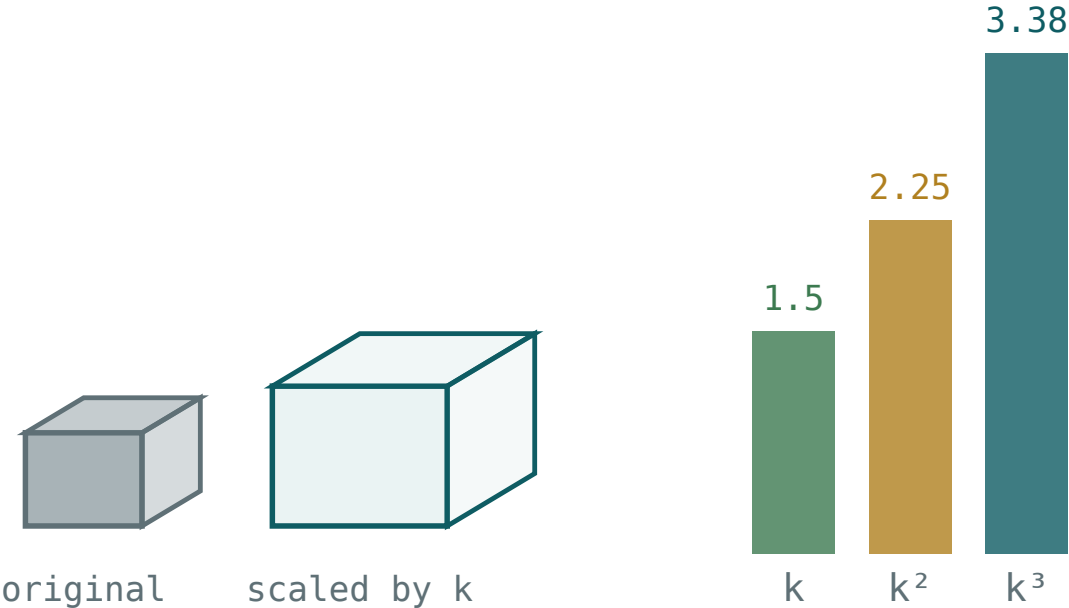
$$② \quad H = \frac{h}{1 - k} = \frac{4}{0.6}$$

$$③ \quad \approx 6.67$$

ANSWER ≈ 6.67 **03** **Interactive model**

Cut a card pyramid parallel to its base and hold up both pieces.

Cutting a pyramid



k 1.5	length × 1.5	area ×k² 2.25	volume ×k³ 3.38
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04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Frustum	Kesik piramida	Усечённая пирамида
Upper base	Yuqori asos	Верхнее основание
Lower base	Quyi asos	Нижнее основание
Slant height of a frustum	Kesik piramida apofemasi	Апофема усечённой пирамиды
Lateral face	Yon yoq	Боковая грань
Similar bases	O'xshash asoslar	Подобные основания
Scale factor	O'xshashlik koeffitsienti	Коэффициент подобия
Complementary cone	To'ldiruvchi piramida	Дополняющая пирамида

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

The lateral faces of a frustum are:

The two bases of a frustum are:

S_{lat} of a frustum:

\frac{1}{2}Pl"/>

\frac{1}{2}(P_1 + P_2)l"/>

(P_1 + P_2)l"/>

Pl"/>

V of a frustum:

$$\frac{h}{2}(B_1 + B_2)$$

$$\frac{h}{3}(B_1 + B_2 + \sqrt{B_1 B_2})$$

$$\frac{1}{3}Bh$$

$$Bh$$

The average-of-bases formula gives an answer that is:

correct

too big

too small

zero

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. S_{lat} with $P_1 = 40$, $P_2 = 16$, $l = 5$

ANSWER 140

2. S_{lat} with $P_1 = 24$, $P_2 = 12$, $l = 4$

ANSWER 72

3. $\sqrt{B_1 B_2}$ for $B_1 = 100$, $B_2 = 16$

ANSWER 40

4. $\sqrt{B_1 B_2}$ for $B_1 = 36$, $B_2 = 9$

ANSWER 18

5. The lateral faces of a frustum are

ANSWER trapezia

6. The bases of a frustum are

ANSWER similar

7. Scale factor for bases 4 and 10

ANSWER 0.4

Medium · the standard question

7 PROBLEMS

1. Bases 10 and 4, slant height 5: total area

ANSWER 256

2. Bases 10 and 4, height 4: the volume

ANSWER 208

3. Height of the original pyramid there

ANSWER ≈ 6.67

4. Bases 6 and 3, height 4: the volume

ANSWER 84

5. Bases 12 and 6, slant height 8: lateral area

ANSWER 288

6. Bases 8 and 4, height 6: the volume

ANSWER 224

7. A frustum cut halfway up: the volume ratio to the whole

ANSWER 7 : 8

Hard · stretch and reason

7 PROBLEMS

1. Bases 9 and 6, slant height 5: total area and volume with $h = 4$

ANSWER $S = 267$; $V \approx 254$

2. A frustum of volume 208 with bases 100 and 16: its height

ANSWER 4

3. Prove $V = \frac{h}{3}(B_1 + B_2 + \sqrt{B_1 B_2})$

ANSWER Subtract the small pyramid from the whole

4. A frustum has bases in ratio 1 : 2. The volume ratio of the frustum to the whole pyramid

ANSWER 7 : 8

5. A bucket: bases 24 and 18 cm, depth 20 cm. Capacity in litres

ANSWER ≈ 6.9 litres

6. A frustum with equal bases is what solid?

ANSWER a prism

7. Show the average formula overestimates

ANSWER $\frac{B_1+B_2}{2} > \frac{B_1+B_2+\sqrt{B_1 B_2}}{3}$ by AM–GM

07 Homework

Homework — 4 tasks

Never average the two bases — use the geometric-mean formula.

1. A frustum of a square pyramid has bases 12 and 6 and slant height 7. Find its total surface area.
2. The same frustum has height 6. Find its volume.
3. Find the height of the pyramid it was cut from.

4. Explain in three sentences why the volume is not $\frac{h}{2}(B_1 + B_2)$.

The cone — surface area and volume

A pyramid whose base is a circle: the same two formulas, with π and a slant height.

LESSON 36–37

2 × 40 MIN

GEOMETRY 11, §17

IGCSE E13.X

LESSON CLOCK

14'

24'

24'

22'

20'

6'

The elements	14 min
The net and the curved area	24 min
The volume	24 min
Working backwards	22 min
Practice	20 min
Homework	6 min

LEARNING OBJECTIVES

- Name the base, axis, height and slant height of a cone.
- Derive the curved surface area from the net.
- Compute the volume of a cone.
- Solve for a missing dimension.

TEXTBOOK

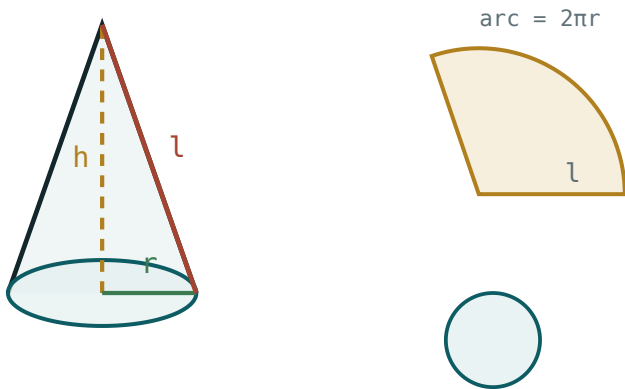
National	pp. 179–190
Cambridge	Core & Extended, pp. 335–342
Workbook	Exercise 17.1

01

Explanation

The elements

A **right circular cone** is swept out when a right triangle turns about one of its legs. Its **axis** joins the apex to the centre of the base circle and is perpendicular to it.



$S = \pi r^2 + \pi r l$ – a circle and a sector

The height, the radius and the slant height form a right triangle. Cut down the slant and it unrolls into a sector.

$$l^2 = h^2 + r^2$$

The **axial section** — the cut through the axis — is an isosceles triangle of base $2r$ and height h , with equal sides l . Every cone question can be reduced to that triangle.

The net and the curved area

Cut the cone down one slant line and unroll it. The curved surface becomes a **sector** of radius l whose arc is the base circumference $2\pi r$. The sector is the fraction $\frac{2\pi r}{2\pi l}$

$= \frac{r}{l}$ of the full circle of radius l :

$$S_{\text{curved}} = \frac{r}{l} \times \pi l^2 = \pi r l$$

$$S_{\text{total}} = \pi r^2 + \pi r l = \pi r(r + l)$$

l , NOT h

The curved surface uses the **slant** height. Using h understates it, and the two are equal only for a flat disc. Compute $l = \sqrt{h^2 + r^2}$ first, every time.

The volume

$$V = \frac{1}{3} \pi r^2 h$$

A cone is exactly one third of the cylinder on the same base with the same height — the same $\frac{1}{3}$ as a pyramid against its prism. Both come from the same integral, $\int x^2 dx$, as Quarter III algebra showed.

SOLID	VOLUME	CURVED / LATERAL AREA
cylinder	$\pi r^2 h$	$2\pi r h$
cone	$\frac{1}{3}\pi r^2 h$	$\pi r l$
prism	Bh	Ph
pyramid	$\frac{1}{3}Bh$	$\frac{1}{2}Pl$

Working backwards

$$h = \frac{3V}{\pi r^2} \quad r = \sqrt{\frac{3V}{\pi h}} \quad l = \frac{S_{\text{curved}}}{\pi r}$$

Example. A cone of volume $100\pi \text{ cm}^3$ and radius 5 cm has $h = \frac{300\pi}{25\pi} = 12 \text{ cm}$, and then $l = 13$ and $S_{\text{curved}} = 65\pi \approx 204 \text{ cm}^2$.

Given the **total** area, the equation is quadratic in r — expand $\pi r(r + l)$ and solve, rejecting the negative root.

02 Worked examples

EXAMPLE 1 A cone has radius 6 cm and height 8 cm. Find its slant height, total surface area and volume.

$$① \quad l = \sqrt{64 + 36} = 10$$

$$② \quad S = \pi(6)(6 + 10) = 96\pi \approx 301.6$$

$$③ \quad V = \frac{1}{3}\pi(36)(8) = 96\pi \approx 301.6$$

A coincidence of these numbers.

ANSWER

$$l = 10; S = 96\pi; V = 96\pi$$

EXAMPLE 2 A cone has volume $100\pi \text{ cm}^3$ and radius 5 cm. Find its height and curved surface area.

$$1 \quad 100\pi = \frac{1}{3}\pi(25)h$$

$$2 \quad h = 12$$

$$3 \quad l = 13$$

$$4 \quad S_{\text{curved}} = \pi(5)(13) = 65\pi \approx 204$$

ANSWER

$$h = 12; 65\pi \approx 204 \text{ cm}^2$$

EXAMPLE 3 A sector of radius 12 and angle 120° is rolled into a cone. Find its radius and height.

$$1 \quad l = 12; \text{arc} = \frac{120}{360} \times 24\pi = 8\pi.$$

$$2 \quad 2\pi r = 8\pi \Rightarrow r = 4$$

$$3 \quad h = \sqrt{144 - 16} = \sqrt{128} \approx 11.3$$

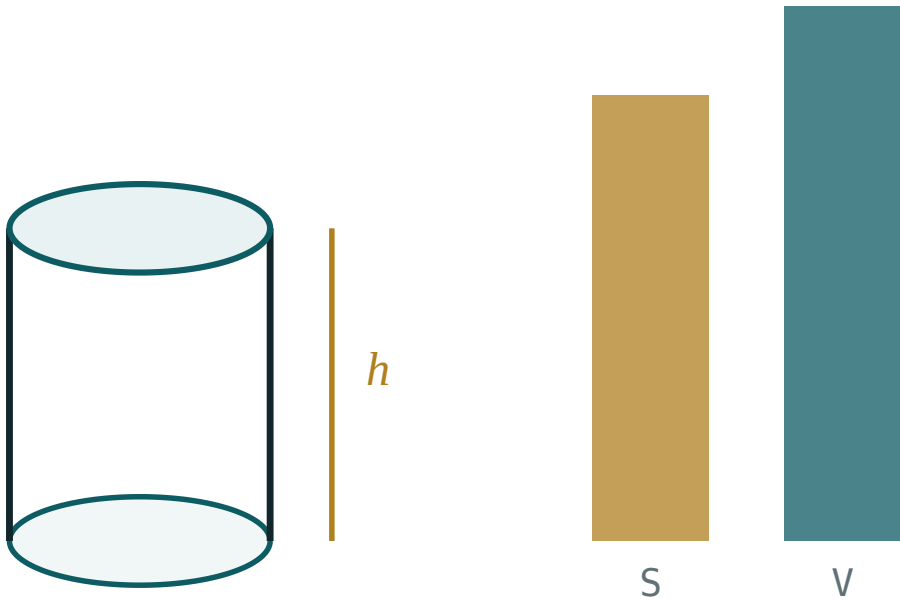
ANSWER

$$r = 4, h \approx 11.3$$

03 Interactive model

Roll a paper sector into a cone and measure the base radius against the arc length.

The cone against the cylinder



base area B 50.3	perimeter P 25.1	total S 251.3	volume V 301.6
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04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Cone	Konus	Конус
Apex	Uchi	Вершина
Axis of a cone	Konus o'qi	Ось конуса
Slant height	Yasovchi	Образующая
Curved surface area	Yon sirt yuzasi	Площадь боковой поверхности
Net of a cone	Konus yoymasi	Развёртка конуса
Axial section	O'qli kesim	Осевое сечение
Right circular cone	To'g'ri doiraviy konus	Прямой круговой конус
Solid of revolution	Aylanish jismi	Тело вращения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The curved surface area of a cone is:

$$\pi r^2$$

$$\pi r l$$

$$\pi r h$$

$$2\pi r l$$

The volume of a cone is:

$$\pi r^2 h$$

$$\frac{1}{3} \pi r^2 h$$

$$\frac{1}{2} \pi r^2 h$$

$$\pi r l$$

l equals:

$$h + r$$

$$\sqrt{h^2 + r^2}$$

$$\sqrt{h^2 - r^2}$$

$$hr$$

The net of a cone is:

a rectangle and two circles

a sector and a circle

two sectors

a triangle

The axial section of a cone is:

a circle

an isosceles triangle

a rectangle

a trapezium

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. l with $r = 3, h = 4$

ANSWER 5

2. l with $r = 5, h = 12$

ANSWER 13

3. Volume with $r = 3, h = 4$

ANSWER 12π

4. Curved area with $r = 3, l = 5$

ANSWER 15π

5. Total area with $r = 3, l = 5$

ANSWER 24π

6. Volume with $r = 6, h = 10$

ANSWER 120π

7. A cone is what fraction of its cylinder?

ANSWER $\frac{1}{3}$

Medium · the standard question

7 PROBLEMS

1. $r = 6, h = 8$: total area and volume

ANSWER 96π each

2. Volume 100π , radius 5: the height

ANSWER 12

3. Same: the curved area

ANSWER 65π

4. Sector radius 12, angle 120° rolled up: r

ANSWER 4

5. Same: the height

ANSWER $\sqrt{128} \approx 11.3$

6. A cone of $r = 7, l = 25$: the height

ANSWER 24

7. A cone of curved area 60π and $r = 5$: l

ANSWER 12

Hard · stretch and reason

7 PROBLEMS

1. A cone of total area 90π and slant height 12: the radius

ANSWER $r = 5$

2. A cone of volume 96π and slant height 10: r and h

ANSWER $r = 6, h = 8$

3. The sector angle for a cone with $r = 4, l = 12$

ANSWER 120°

4. A cone whose axial section is equilateral, $r = 5$: its volume

ANSWER $\frac{125\pi\sqrt{3}}{3} \approx 226.7$

5. A cone of maximum volume from a sector of radius 12: the angle

ANSWER $\approx 294^\circ$

6. Two similar cones with volumes 8 and 27: the area ratio

ANSWER $4 : 9$

7. A cone is inscribed in a cube of edge 12: its volume

ANSWER 144π

07 Homework

Homework — 5 tasks

Compute l before any surface-area formula.

1. A cone has radius 9 cm and height 12 cm. Find its slant height, total surface area and volume.
2. A cone has volume 147π cm³ and radius 7 cm. Find its height and curved surface area.
3. A sector of radius 15 and angle 144° is rolled into a cone. Find its radius, height and volume.
4. A cone has total surface area 144π and slant height 15. Find its radius.

5. Explain in three sentences why the curved surface area uses l and not h .

Arc, sector and the net of a cone

Cambridge insert: the radian formulas of Grade 10, used to build a cone from a flat sheet.

LESSON 38

1 × 40 MIN

GEOMETRY 11, §17 (EXTENSION)

P1 · 4.2–4.3

LESSON CLOCK

10'

14'

12'

6'

3'

The link between the two circles	10 min
Finding the angle	14 min
Waste	12 min
A design task	6 min
Homework	3 min

LEARNING OBJECTIVES

- Use $s = r\theta$ and $A = \frac{1}{2}r^2\theta$ in a practical construction.
- Find the sector angle needed to build a cone of given dimensions.
- Compute the wasted material when a sector is cut from a sheet.
- Design a cone to a specification.

TEXTBOOK

National	pp. 191–194
Cambridge	Pure Mathematics 1, pp. 86–95
Workbook	P1 Exercise 4B

01

Explanation

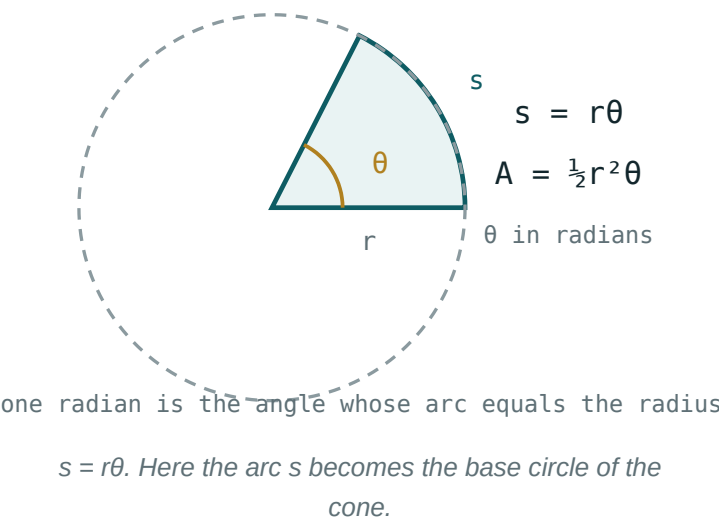
The link between the two circles

THE ONE EQUATION

The arc of the sector becomes the circumference of the base:

$$l\theta = 2\pi r$$

with θ in radians, l the sector radius (the cone's slant height) and r the cone's base radius.



Everything else follows: $\theta = \frac{2\pi r}{l}$ in radians, or $\frac{360r}{l}$ degrees.

Finding the angle

L	R	Θ	IN DEGREES
12	4	$\frac{2\pi}{3}$	120°
10	5	π	180°
15	6	$\frac{4\pi}{5}$	144°
l	l	2π	360°

The last row is the flat disc: when $r = l$ the cone has no height at all.

Check the areas agree: the sector area $\frac{1}{2}l^2\theta = \frac{1}{2}l^2 \cdot \frac{2\pi r}{l} = \pi rl$ — the curved surface area of the last lesson, derived a second way.

Waste, and a design task

A sector is cut from a square or a circular sheet; the rest is waste. The proportion wasted is a real design consideration:

waste fraction from a full disc = $1 - \frac{\theta}{2\pi} = 1 - \frac{r}{l}$

Example. A cone with $r = 4$, $l = 12$ uses $\frac{1}{3}$ of the disc of radius 12, so two thirds of the material is wasted. A shallower cone — r closer to l — wastes far less.

THE DESIGN TASK, IN PAIRS

Design a paper cone to hold exactly 200 ml (200 cm^3). Choose r and h , compute l and θ , draw the sector, cut and build it. Then measure what it actually holds and explain the difference.

02 Worked examples

EXAMPLE 1 A cone has $r = 5$ and $l = 15$. Find the sector angle in degrees.

$$\textcircled{1} \quad \theta = \frac{2\pi r}{l} = \frac{10\pi}{15} = \frac{2\pi}{3}$$

$$\textcircled{2} \quad \text{In degrees: } \frac{360 \times 5}{15}$$

$$\textcircled{3} \quad = 120^\circ$$

ANSWER

120°

EXAMPLE 2 A sector of radius 20 and angle 216° is rolled up. Find the cone's radius and volume.

$$① \quad r = \frac{216}{360} \times 20 = 12$$

$$② \quad h = \sqrt{400 - 144} = 16$$

$$③ \quad V = \frac{1}{3} \pi (144)(16)$$

$$④ \quad = 768\pi \approx 2413$$

ANSWER

$$r = 12, V = 768\pi \approx 2413$$

EXAMPLE 3 Design a cone of volume 200 cm^3 with $r = 5$. Find h , l and θ .

$$① \quad 200 = \frac{1}{3} \pi (25)h \Rightarrow h = \frac{24}{\pi} \approx 7.64$$

$$② \quad l = \sqrt{25 + 58.3} \approx 9.13$$

$$③ \quad \theta = \frac{360 \times 5}{9.13} \approx 197^\circ$$

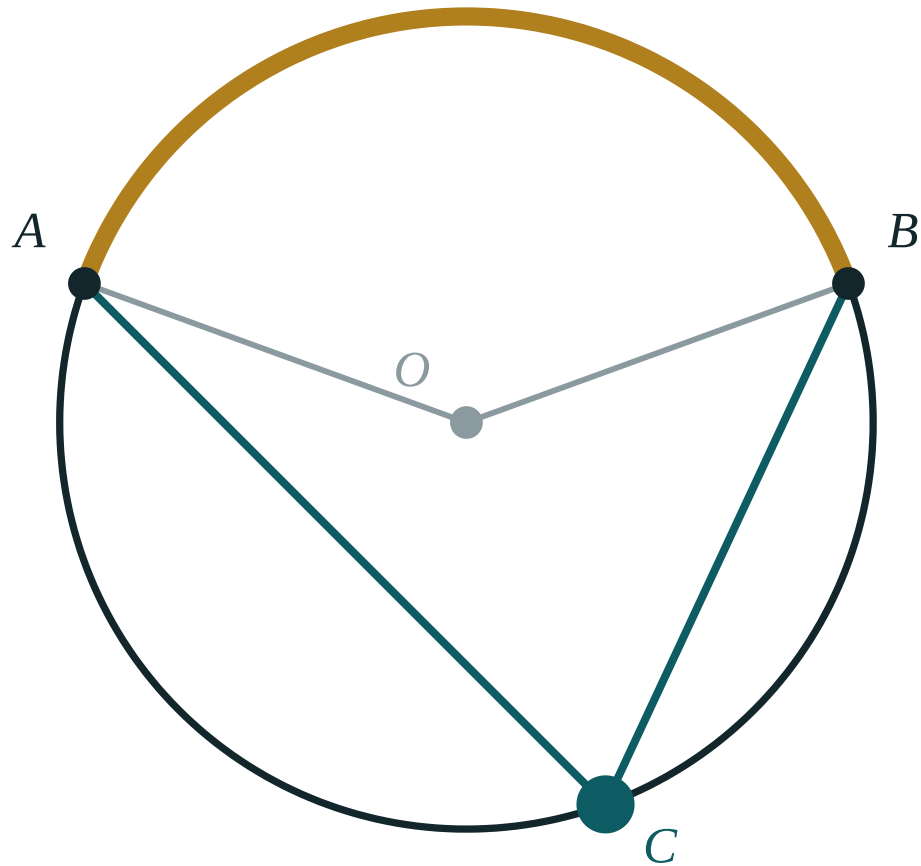
ANSWER

$$h \approx 7.6, l \approx 9.1, \theta \approx 197^\circ$$

03 Interactive model

Cut two sectors of the same radius but different angles and compare the cones.

Sector to cone

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sector	Sektor	Сектор
Arc length	Yoy uzunligi	Длина дуги
Sector angle	Sektor burchagi	Угол сектора
Radian	Radian	Радян
Net	Yoyma	Развёртка
Waste material	Chiqindi material	Отходы материала
Circumference	Aylana uzunligi	Длина окружности
Design specification	Loyiha talabi	Техническое задание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

The arc of the sector becomes:

the slant height

the base circumference

the height

the axis

θ in radians is:

$$\frac{r}{l}$$

$$\frac{2\pi r}{l}$$

$$\frac{l}{r}$$

$$2\pi l$$

$l = 12, r = 4$ gives θ :

$$60^\circ$$

$$120^\circ$$

$$180^\circ$$

$$240^\circ$$

When $r = l$ the cone is:

very tall

a flat disc

a sphere

impossible

The sector area equals:

πr^2

$\pi r l$

$2\pi r l$

πl^2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. θ for $l = 10, r = 5$

ANSWER 180°

2. θ for $l = 12, r = 4$

ANSWER 120°

3. θ for $l = 15, r = 6$

ANSWER 144°

4. r from $l = 20, \theta = 90^\circ$

ANSWER 5

5. r from $l = 18, \theta = 180^\circ$

ANSWER 9

6. The arc becomes

ANSWER the base circumference

7. Sector area equals

ANSWER $\pi r l$

Medium · the standard question

7 PROBLEMS

1. Sector radius 20, angle 216° : the cone's radius

ANSWER 12

2. Same: the height and volume

ANSWER $16, 768\pi$

3. Cone of $r = 5, l = 15$: the angle

ANSWER 120°

4. Waste fraction for $r = 4, l = 12$

ANSWER $\frac{2}{3}$

5. Waste fraction for $r = 9, l = 10$

ANSWER 0.1

6. Cone of volume 200 with $r = 5$: h

ANSWER ≈ 7.64

7. Same: the sector angle

ANSWER $\approx 197^\circ$

Hard · stretch and reason

7 PROBLEMS

1. A cone of maximum volume from a disc of radius R : the sector angle

ANSWER $\approx 294^\circ$

2. That maximum volume for $R = 12$

ANSWER ≈ 836

3. A cone with $\theta = 90^\circ$ and volume 200: find l

ANSWER ≈ 11.7

4. Two cones from one disc of radius 10, no waste: their radii

ANSWER 5 each if the disc is halved

5. A cone of $r = 6$, $h = 8$: the sector angle and the waste from a disc of radius l

ANSWER $216^\circ, 40\%$

6. Design a cone of volume 500 and $\theta = 180^\circ$

ANSWER $r \approx 6.2, l \approx 12.4$

7. Why does a shallower cone waste less material?

ANSWER r is nearer l , so the sector is nearer the full disc

07 Homework

Homework — 4 tasks

Bring the built paper cone and its measured capacity.

1. A cone has $r = 8$ and $l = 20$. Find the sector angle in degrees and in radians.
2. A sector of radius 24 and angle 150° is rolled up. Find the cone's radius, height and volume.
3. Design and build a paper cone of volume 200 cm^3 , giving r , h , l and θ .
4. Measure what your cone actually holds and explain in two sentences why it differs from 200 ml.

The frustum of a cone

The bucket, the lampshade and the flowerpot — and the two formulas that measure them.

LESSON 39

1 × 40 MIN

GEOMETRY 11, §18

EXTENSION

LESSON CLOCK

8'

14'

14'

6'

3'

The solid	8 min
Surface area	14 min
Volume	14 min
Practical problems	6 min
Homework	3 min

LEARNING OBJECTIVES

- Describe the elements of a frustum of a cone.
- Compute its curved and total surface area.
- Compute its volume.
- Solve a practical problem about a bucket or a lampshade.

TEXTBOOK

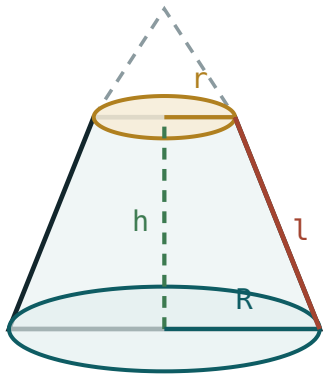
National	pp. 195–200
Cambridge	Core & Extended, pp. 343–346
Workbook	Exercise 18.1

01

Explanation

The solid

Cut a cone with a plane parallel to the base and remove the small cone. What remains is a **frustum of a cone** — the shape of every bucket, lampshade and flowerpot.



$$V = \frac{1}{3}\pi h(R^2 + Rr + r^2) \cdot S_{\text{lat}} = \pi l(R + r)$$

The removed cone shown dashed. R and r are the two radii; l is the slant height of the frustum.

$$l^2 = h^2 + (R - r)^2$$

The axial section is a trapezium with parallel sides $2R$ and $2r$, height h and slanting sides l . The $R - r$ in the formula is the horizontal step from one radius to the other.

Surface area

$$S_{\text{curved}} = \pi l(R + r) \quad S_{\text{total}} = \pi l(R + r) + \pi R^2 + \pi r^2$$

It is the big cone's curved area minus the small one's, simplified — the same construction as the pyramid frustum, with $\frac{1}{2}(P_1 + P_2)l$ becoming $\pi(R + r)l$.

A BUCKET HAS ONE END, NOT TWO

A bucket needs $\pi l(R + r) + \pi r^2$ — the curved surface and the **base** only. Read the object before choosing which circles to include.

Volume

$$V = \frac{\pi h}{3}(R^2 + Rr + r^2)$$

The same shape of formula as the pyramid frustum, with $\sqrt{B_1 B_2}$ becoming πRr . With $r = R$ it gives the cylinder $\pi R^2 h$; with $r = 0$ it gives the cone $\frac{1}{3}\pi R^2 h$. Both checks are worth doing once.

Example. A bucket of radii 12 and 9 cm and depth 20 cm:

$$V = \frac{20\pi}{3}(144 + 108 + 81) = \frac{20\pi}{3}(333) = 2220\pi \approx 6975 \text{ cm}^3$$

That is about 7 litres — a realistic bucket.

Practical problems

OBJECT	WHAT IS WANTED	WHICH FORMULA
a bucket	capacity	$\frac{\pi h}{3}(R^2 + Rr + r^2)$
a lampshade	material	$\pi l(R + r)$
a flowerpot	capacity and the material for the sides and base	both
a paper cup	the flat sector-ring to cut	the net

THE CHECK THAT CATCHES ERRORS

Put $r = R$ into any frustum formula: it must give the cylinder. Put $r = 0$: it must give the cone. A formula that fails either check has been mis-copied.

02 Worked examples

EXAMPLE 1 A bucket has radii 12 and 9 cm and depth 20 cm. Find its capacity in litres.

1 $V = \frac{20\pi}{3}(144 + 108 + 81)$

2 $= \frac{20\pi}{3} \times 333 = 2220\pi$

3 $\approx 6975 \text{ cm}^3$

4 $\approx 7.0 \text{ litres.}$

ANSWER

$\approx 7.0 \text{ litres}$

EXAMPLE 2 The same bucket: find the area of metal needed (curved surface and base).

$$① \quad l = \sqrt{400 + 9} = \sqrt{409} \approx 20.2$$

$$② \quad S_{\text{curved}} = \pi(20.2)(21) \approx 1333$$

$$③ \quad \text{base} = \pi(81) \approx 254$$

$$④ \quad \approx 1587 \text{ cm}^2.$$

ANSWER

$$\approx 1590 \text{ cm}^2$$

EXAMPLE 3 A frustum has radii 6 and 3 and slant height 5. Find its height and volume.

$$① \quad h = \sqrt{25 - 9} = 4$$

$$② \quad V = \frac{4\pi}{3}(36 + 18 + 9)$$

$$③ \quad = \frac{4\pi}{3} \times 63 = 84\pi$$

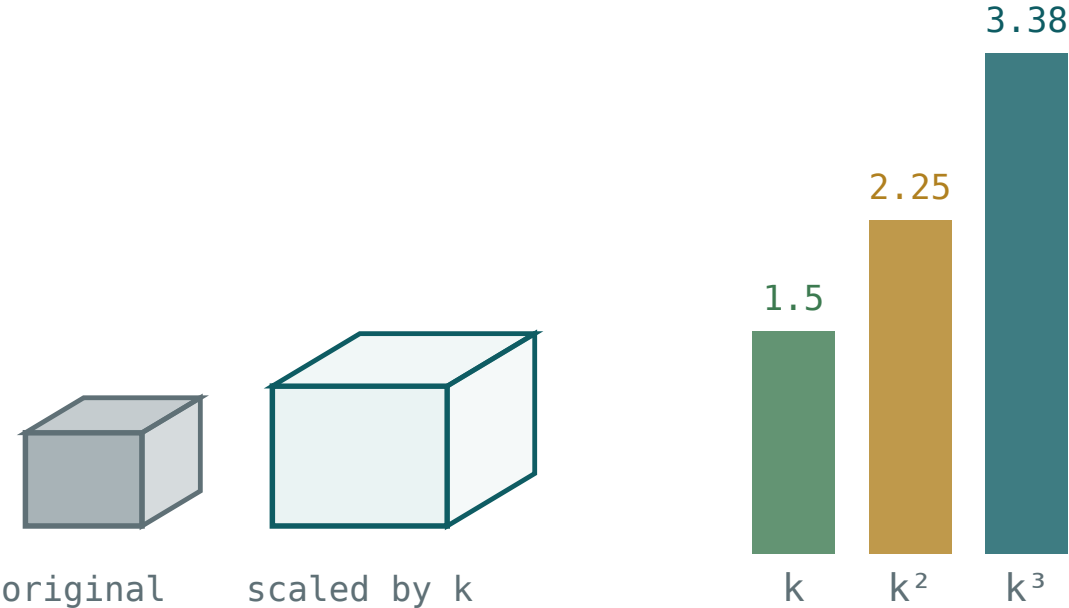
ANSWER

$$h = 4, V = 84\pi \approx 264$$

03 Interactive model

Bring a real bucket and measure its two radii and depth.

Cutting a cone



k 1.5	length × 1.5	area ×k² 2.25	volume ×k³ 3.38
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04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Frustum of a cone	Kesik konus	Усечённый конус
Upper radius	Yuqori radius	Верхний радиус
Lower radius	Quyi radius	Нижний радиус
Slant height	Yasovchi	Образующая
Curved surface	Yon sirt	Боковая поверхность
Capacity	Sig'im	Вместимость
Complementary cone	To'ldiruvchi konus	Дополняющий конус
Axial section	O'qli kesim	Осевое сечение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05

Quick check
Check yourself

l^2 for a frustum equals:

$$h^2 + R^2$$

$$h^2 + (R - r)^2$$

$$h^2 + (R + r)^2$$

$$R^2 - r^2$$

S_{curved} of a frustum:

$$\pi r l$$

$$\pi l(R + r)$$

$$\pi l(R - r)$$

$$2\pi r l$$

V of a frustum:

$$\frac{\pi h}{2}(R^2 + r^2)$$

$$\frac{\pi h}{3}(R^2 + Rr + r^2)$$

$$\frac{1}{3}\pi R^2 h$$

$$\pi R^2 h$$

Setting $r = R$ gives:

a cone

a cylinder

a sphere

nothing

Setting $r = 0$ gives:

a cone

a cylinder

a sphere

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. l with $h = 4$, $R = 6$, $r = 3$

ANSWER 5

2. l with $h = 12$, $R = 10$, $r = 5$

ANSWER 13

3. S_{curved} with $l = 5$, $R = 6$, $r = 3$

ANSWER 45π

4. V with $h = 4$, $R = 6$, $r = 3$

ANSWER 84π

5. Setting $r = R$ gives

ANSWER a cylinder

6. Setting $r = 0$ gives

ANSWER a cone

7. Axial section of a frustum

ANSWER a trapezium

Medium · the standard question

7 PROBLEMS

1. Bucket $R = 12, r = 9, h = 20$: capacity in litres

ANSWER ≈ 7.0

2. Same: metal for the sides and base

ANSWER $\approx 1590 \text{ cm}^2$

3. $R = 10, r = 4, h = 8$: the volume

ANSWER $\frac{8\pi}{3} \times 156 = 416\pi$

4. Same: the slant height

ANSWER 10

5. Same: the curved area

ANSWER 140π

6. A lampshade $R = 20, r = 12, h = 15$: material

ANSWER $l = 17$, curved area 544π

7. A flowerpot $R = 15, r = 10, h = 18$: capacity in litres

ANSWER ≈ 8.9

Hard · stretch and reason

7 PROBLEMS

1. A frustum of volume 84π with $R = 6$, $r = 3$: its height

ANSWER 4

2. A bucket holds 10 litres with $R = 14$, $r = 11$: its depth

ANSWER ≈ 20.6 cm

3. Show V reduces to the cone when $r = 0$

ANSWER The other two terms vanish

4. A frustum cut halfway up a cone: the volume ratio frustum : whole

ANSWER 7 : 8

5. A frustum with $R = 2r$ and height h : its volume in terms of r , h

ANSWER $\frac{7\pi r^2 h}{3}$

6. A paper cup $R = 4$, $r = 2.5$, $h = 9$: capacity in ml

ANSWER ≈ 313

7. Two similar frustums with heights 3 and 5: the volume ratio

ANSWER 27 : 125

07 Homework

Homework — 4 tasks

Check every formula by setting $r = R$ and $r = 0$.

1. A bucket has radii 14 and 10 cm and depth 24 cm. Find its capacity in litres.
2. The same bucket: find the area of metal for the curved surface and the base.
3. A frustum has radii 8 and 5 and slant height 5. Find its height, curved area and volume.
4. Show that the frustum volume formula gives the cylinder when $r = R$ and the cone when $r = 0$.

Similar solids and the volume scale factor

Cambridge insert: one length ratio, three consequences — used on real models and real costs.

LESSON 40

1 × 40 MIN

GEOMETRY 11, §18 (EXTENSION)

IGCSE E21.1–21.2

LESSON CLOCK

8'

14'

12'

8'

3'

The three rules	8 min
Working backwards	14 min
Applications	12 min
Why size matters	8 min
Homework	3 min

LEARNING OBJECTIVES

- Apply the k , k^2 and k^3 rules to similar solids.
- Recover the scale factor from an area or a volume ratio.
- Solve problems on models, packaging and costs.
- Explain why large animals are proportionally thicker.

TEXTBOOK

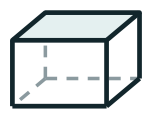
National	pp. 201–204
Cambridge	Core & Extended, pp. 226–234
Workbook	IGCSE Exercise 21.2

01

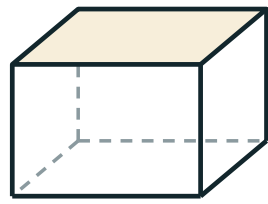
Explanation

The three rules

$$\frac{L_2}{L_1} = k \cdot \frac{S_2}{S_1} = k^2 \cdot \frac{V_2}{V_1} = k^3$$



$k = 1$



$k = 2$

lengths $\times k$ · areas $\times k^2$ · volumes $\times k^3$

Double every length: four times the surface, eight times the volume.

Two solids are similar when one is an enlargement of the other. All cubes are similar; all spheres are similar; two cones are similar exactly when $\frac{r}{h}$ is the same for both.

Working backwards

$$k = \sqrt{\frac{S_2}{S_1}} \quad k = \sqrt[3]{\frac{V_2}{V_1}}$$

GIVEN	<i>K</i>	OTHER FACTORS
volumes 8 : 27	$\frac{2}{3}$	areas 4 : 9
areas 16 : 25	$\frac{4}{5}$	volumes 64 : 125
lengths 1 : 50	$\frac{1}{50}$	areas 1 : 2500, volumes 1 : 125 000

Applications

QUESTION	WHICH FACTOR
how much paint?	k^2
how much liquid?	k^3
how much wire round the edge?	k
how much does it weigh?	k^3
how much fabric to cover it?	k^2

Example. A model at 1 : 20 needs 0.15 litres of paint. The real object needs $0.15 \times 400 = 60$ litres — not 3.

PAINT IS AN AREA, NOT A VOLUME

A coat of paint is a thin layer over a surface, so it scales with k^2 . Filling a tank scales with k^3 . Reading which is asked is the whole question.

Why size matters

As a solid grows, its volume outruns its surface:

$$\frac{S}{V} \text{ scales as } \frac{k^2}{k^3} = \frac{1}{k}$$

THREE CONSEQUENCES

A large animal has proportionally **less** skin per kilogram, so it loses heat more slowly — which is why elephants have big ears and mice do not.

Its mass grows as k^3 but its bone cross-section only as k^2 , so it needs proportionally thicker legs.

Small objects cool, dry and dissolve faster — the reason for chopping food and grinding ore.

Doubling an animal's height multiplies its weight by 8 and its bone strength by only 4.

The scaled-up animal would break its own legs — a real constraint, first noticed by Galileo.

02 Worked examples

EXAMPLE 1 Two similar cones have volumes 54 and 128 cm³. Find the ratio of their surface areas.

$$\textcircled{1} \quad k^3 = \frac{128}{54} = \frac{64}{27}$$

$$\textcircled{2} \quad k = \frac{4}{3}$$

$$\textcircled{3} \quad k^2 = \frac{16}{9}$$

ANSWER

9 : 16

EXAMPLE 2 A model car is $1 : 24$. It needs 12 cm^2 of paint. The real car needs?

① $k = 24$, so $k^2 = 576$.

② $12 \times 576 = 6912 \text{ cm}^2$.

③ $= 0.69 \text{ m}^2$.

ANSWER

$\approx 0.69 \text{ m}^2$

EXAMPLE 3 Two similar jugs hold 0.5 and 1.5 litres. Their heights are in what ratio?

① $k^3 = 3$

② $k = \sqrt[3]{3} \approx 1.442$

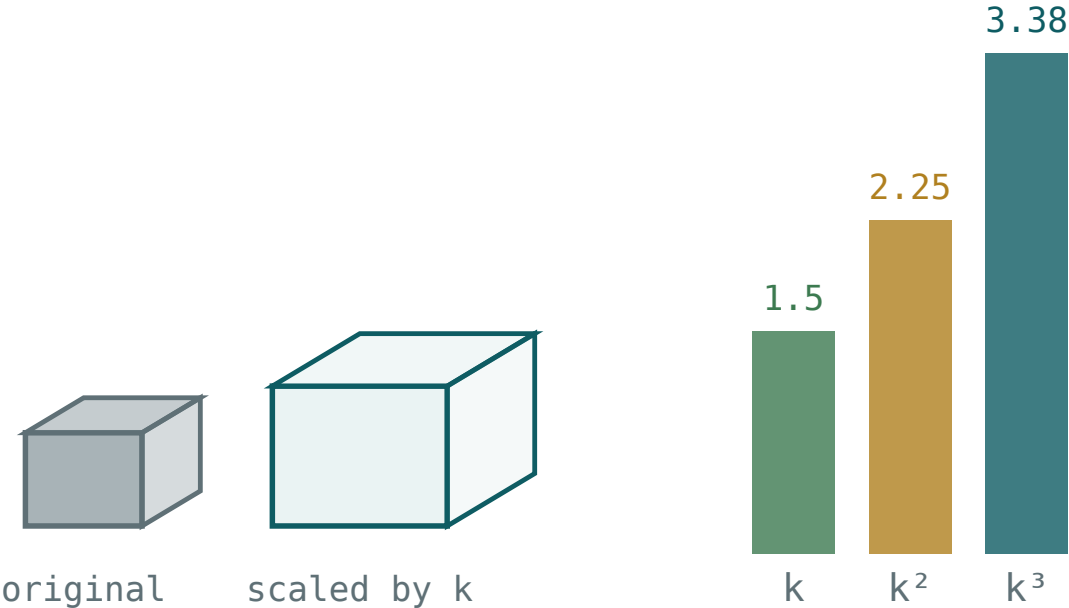
ANSWER

$1 : 1.44$

03 Interactive model

Compare two similar containers and predict, then measure, the capacity ratio.

Scaling a solid



k 1.5	length × 1.5	area ×k² 2.25	volume ×k³ 3.38
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04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Similar solids	O'xshash jismlar	Подобные тела
Scale factor	O'xshashlik koeffitsienti	Коэффициент подобия
Area factor	Yuza koeffitsienti	Коэффициент площади
Volume factor	Hajm koeffitsienti	Коэффициент объёма
Model	Model	Модель
Corresponding lengths	Mos uzunliklar	Соответственные длины
Cube root	Kub ildiz	Кубический корень
Surface-area-to-volume ratio	Sirt va hajm nisbati	Отношение поверхности к объёму

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Doubling all lengths multiplies the volume by:

And the surface area by:

Volumes $8 : 27$ gives lengths:

Paint scales with:

k

k^2

k^3

nothing

S V scales as:

k

$\frac{1}{k}$

k^2

k^3

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $k = 3$: area factor

ANSWER 9

2. $k = 3$: volume factor

ANSWER 27

3. $k = 0.5$: volume factor

ANSWER 0.125

4. Volumes 27 : 64: lengths

ANSWER 3 : 4

5. Areas 9 : 25: lengths

ANSWER 3 : 5

6. Paint scales with

ANSWER k^2

7. Liquid scales with

ANSWER k^3

Medium · the standard question

7 PROBLEMS

1. Volumes 54 and 128: the area ratio

ANSWER 9 : 16

2. Model 1 : 24 needing 12 cm² of paint: the real area

ANSWER ≈ 0.69 m²

3. Jugs of 0.5 and 1.5 litres: the height ratio

ANSWER 1 : 1.44

4. Model 1 : 50: the volume ratio

ANSWER 1 : 125 000

5. A statue 1 : 6 needs 0.5 litres of paint. The real one

ANSWER 18 litres

6. Two similar tins, heights 8 and 12; smaller holds 320 ml

ANSWER 1080 ml

7. Areas 49 : 81: the volume ratio

ANSWER 343 : 729

Hard · stretch and reason

7 PROBLEMS

1. Two similar solids differ in volume by 111 with $k = \frac{4}{3}$

ANSWER 81 and 192

2. A cone cut halfway up: the volume ratio of the parts

ANSWER 1 : 7

3. A 1 : 200 model bridge with deck area 80 cm²: the real area

ANSWER 320 m²

4. Why can a mouse survive a fall an elephant cannot?

ANSWER Its surface-to-mass ratio is far larger

5. A doubled animal: weight and bone strength factors

ANSWER 8 and 4

6. Two similar containers hold 2 and 6.75 litres. The height ratio

ANSWER 2 : 3

7. A sugar cube dissolves faster when crushed. Why?

ANSWER Crushing multiplies the surface area at the same volume

07 Homework

Homework — 4 tasks

State k explicitly before using k^2 or k^3 .

- Two similar cones have volumes 40 and 135 cm³. Find the ratio of their heights and of their surface areas.
- A model aeroplane is 1 : 72. Its wing area is 60 cm². Find the real wing area in m².
- Two similar bottles hold 330 and 1000 ml. Find the ratio of their heights.

4. Explain, using k^2 and k^3 , why a doubled animal could not stand.

The sphere and its sections; inscribed and circumscribed solids

Every plane section is a circle — and that single fact settles every sphere problem.

LESSON 41–42

2 × 40 MIN

GEOMETRY 11, §19

IGCSE E13.X

LESSON CLOCK

12'

24'

24'

24'

20'

6'

Sphere and ball	12 min
Plane sections	24 min
Tangent planes	24 min
Inscribed and circumscribed	24 min
Practice	20 min
Homework	6 min

LEARNING OBJECTIVES

- Define a sphere and a ball, and name their elements.
- Compute the radius of a plane section from its distance to the centre.
- Recognise great and small circles.
- Solve problems on solids inscribed in or circumscribed about a sphere.

TEXTBOOK

National	pp. 205–216
Cambridge	Core & Extended, pp. 347–354
Workbook	Exercise 19.1

01

Explanation

Sphere and ball

TWO DIFFERENT OBJECTS

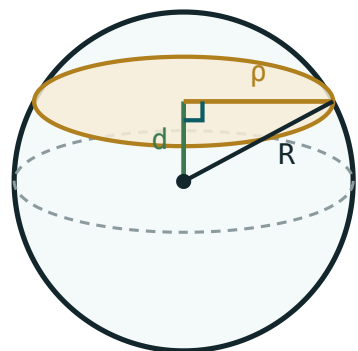
A **sphere** is the **surface** — all points at distance R from the centre. A **ball** is the solid — all points at distance $\leq R$. Surface area belongs to the sphere; volume belongs to the ball.

In coordinates (Quarter I) the sphere of centre (a, b, c) and radius R is

$$(x - a)^2 + (y - b)^2 + (z - c)^2 = R^2$$

The sphere has infinitely many planes and axes of symmetry, and one centre — the most symmetric solid there is.

Plane sections



the section is a circle of radius ρ
 $\rho^2 = R^2 - d^2$

A plane at distance d from the centre cuts a circle of radius ρ , with $\rho^2 = R^2 - d^2$.

$\rho^2 = R^2 - d^2$

D	SECTION	NAME
0	a circle of radius R	a great circle
$0 < d < R$	a circle of radius $\sqrt{R^2 - d^2}$	a small circle
$d = R$	a single point	a tangent plane
$d > R$	nothing	the plane misses the sphere

The great circles are the largest sections, and the shortest path between two points on a sphere runs along one — which is why aircraft routes look curved on a flat map.

Tangent planes

TANGENT PLANE

A plane at distance exactly R from the centre touches the sphere at one point, and is **perpendicular** to the radius drawn to that point.

That perpendicularity is the space version of the circle theorem of Grade 8, and it is proved the same way: any other point of the plane is further from the centre than the foot of the perpendicular, so it lies outside the sphere.

Inscribed and circumscribed

SITUATION	RELATION
sphere inscribed in a cube of edge a	$R = \frac{a}{2}$
sphere circumscribed about a cube of edge a	$R = \frac{a\sqrt{3}}{2}$
sphere inscribed in a cylinder of radius r , height $2r$	$R = r$
sphere circumscribed about a cuboid $a \times b \times c$	$R = \frac{\sqrt{a^2 + b^2 + c^2}}{2}$

THE TWO QUESTIONS TO ASK

Inscribed: the sphere touches the faces, so its diameter is the smallest dimension. **Circumscribed:** the sphere passes through the vertices, so its diameter is the space diagonal.

Archimedes' result. A sphere inscribed in a cylinder of the same radius and height $2r$ has exactly $\frac{2}{3}$ of the cylinder's volume — and exactly $\frac{2}{3}$ of its total surface area. Archimedes had it carved on his tomb.

02

Worked examples

EXAMPLE 1 A plane is 8 cm from the centre of a sphere of radius 17 cm. Find the radius of the section.

1

$\rho^2 = 289 - 64 = 225$

2

$\rho = 15$

ANSWER

15 cm

EXAMPLE 2 A sphere is inscribed in a cube of edge 12. Find its radius. Then find the radius of the sphere through the cube's vertices.

① Inscribed: touches the faces, $2R = 12$.
 $R = 6$

② Circumscribed: passes through the vertices.
 $2R = 12\sqrt{3}$

③ $R = 6\sqrt{3} \approx 10.4$

ANSWER

6 and $6\sqrt{3}$

EXAMPLE 3 A section of a sphere has radius 12 and is 9 from the centre. Find the sphere's radius and its surface area.

① $R^2 = 144 + 81 = 225$

② $R = 15$

③ $S = 4\pi(225) = 900\pi \approx 2827$

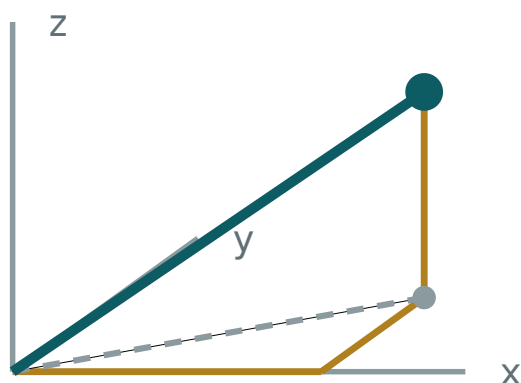
ANSWER

$R = 15$; $900\pi \approx 2830$

03 Interactive model

Cut an orange at different heights and compare the circles.

Cutting a sphere



x 3

y 2

z 2

OM 4.12

base diagonal 3.61

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sphere	Sfera	Сфера
Ball	Shar	Шар
Great circle	Katta aylana	Большая окружность
Small circle	Kichik aylana	Малая окружность
Section of a sphere	Sfera kesimi	Сечение сферы
Tangent plane	Urinma tekislik	Касательная плоскость
Inscribed sphere	Ichki chizilgan sfera	Вписанная сфера
Circumscribed sphere	Tashqi chizilgan sfera	Описанная сфера
Diameter	Diametr	Диаметр

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Every plane section of a sphere is:

an ellipse

a circle

a parabola

a point

ρ^2 equals:

$R^2 + d^2$

$R^2 - d^2$

$d^2 - R^2$

Rd

A great circle has d equal to:

R

0

$\frac{R}{2}$

$2R$

A sphere inscribed in a cube of edge a has radius:

$$a$$

$$\frac{a}{2}$$

$$\frac{a\sqrt{3}}{2}$$

$$a\sqrt{2}$$

A sphere through a cube's vertices has radius:

$$\frac{a}{2}$$

$$\frac{a\sqrt{2}}{2}$$

$$\frac{a\sqrt{3}}{2}$$

$$a$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. ρ with $R = 5, d = 3$

ANSWER 4

2. ρ with $R = 13, d = 5$

ANSWER 12

3. ρ with $R = 17, d = 8$

ANSWER 15

4. Section at $d = 0$ is called

ANSWER a great circle

5. Section at $d = R$ is

ANSWER a single point

6. Sphere inscribed in a cube of edge 10: R

ANSWER 5

7. Sphere round a cube of edge 10: R

ANSWER $5\sqrt{3}$

Medium · the standard question

7 PROBLEMS

1. Section radius 12 at $d = 9$: the sphere's radius

ANSWER 15

2. Same: the surface area

ANSWER 900π

3. Sphere round a cuboid $3 \times 4 \times 12$: its radius

ANSWER 6.5

4. Sphere inscribed in a cylinder $r = 6$, $h = 12$: R

ANSWER 6

5. A plane 10 from the centre of a sphere of radius 26: the section radius

ANSWER 24

6. A section of radius 8 in a sphere of radius 10: its distance

ANSWER 6

7. A plane 15 from a sphere of radius 12 cuts

ANSWER nothing

Hard · stretch and reason

7 PROBLEMS

1. A sphere of radius 25; two parallel planes on the same side at 7 and 20. The section radii

ANSWER 24 and 15

2. A cone of base radius 6 and height 8 inscribed in a sphere: the sphere's radius

ANSWER 6.25

3. Archimedes: the sphere's volume as a fraction of its cylinder

ANSWER $\frac{2}{3}$

4. And the surface areas

ANSWER $\frac{2}{3}$ as well

5. A sphere touches all 12 edges of a cube of edge a : its radius

ANSWER $\frac{a\sqrt{2}}{2}$

6. Two spheres of radii 5 and 3 with centres 6 apart: the radius of the common circle

ANSWER ≈ 2.99

7. Prove a tangent plane is perpendicular to the radius at the point of contact

ANSWER Any other point is further from the centre

07 Homework

Homework — 5 tasks

Draw the right triangle with legs d and p and hypotenuse R every time.

1. A plane is 9 cm from the centre of a sphere of radius 41 cm. Find the section radius.
2. A section of radius 20 lies 21 from the centre. Find the sphere's radius and surface area.
3. Find the radius of the sphere inscribed in, and of the sphere circumscribed about, a cube of edge 8.

4. Find the radius of the sphere through the vertices of a $6 \times 8 \times 24$ cuboid.
5. Explain in three sentences the difference between a sphere and a ball.

Trigonometry applied to solids of revolution

Cambridge insert: the axial section is a plane triangle, and every cone problem lives inside it.

LESSON 43–44

2 × 40 MIN

GEOMETRY 11, §19 (EXTENSION)

IGCSE E6.6

LESSON CLOCK

14'

24'

24'

22'

20'

6'

The axial section	14 min
The semi-vertical angle	24 min
Frustums and inscribed solids	24 min
Combined problems	22 min
Practice	20 min
Homework	6 min

LEARNING OBJECTIVES

- Extract the axial section of a cone or a frustum and solve it.
- Find the semi-vertical angle of a cone.
- Use the sine and cosine rules inside a solid of revolution.
- Combine a trigonometric result with a volume formula.

TEXTBOOK

National	pp. 217–222
Cambridge	Core & Extended, pp. 300–312
Workbook	IGCSE Exercise 6.6

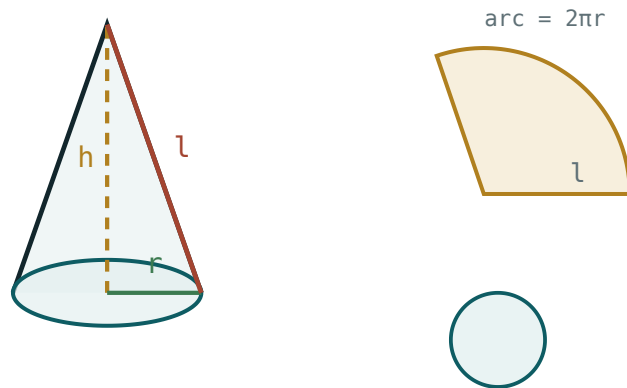
01

Explanation

The axial section

THE ONE MOVE

Cut the solid through its axis. A cone gives an isosceles triangle of base $2r$ and height h ; a cylinder gives a rectangle $2r$ by h ; a frustum gives a trapezium. Draw that flat figure separately and solve it as a plane problem.



$$S = \pi r^2 + \pi r l - \text{a circle and a sector}$$

The right triangle inside the cone: r , h and l .

Every 3-D question about a solid of revolution reduces to this figure, exactly as every cuboid question in Grade 10 reduced to one extracted triangle.

The semi-vertical angle

The angle α between the axis and a slant line is the **semi-vertical angle**; the full angle at the apex is 2α .

$$\tan \alpha = \frac{r}{h} \quad \sin \alpha = \frac{r}{l} \quad \cos \alpha = \frac{h}{l}$$

The angle β the slant makes with the **base** is the complement: $\beta = 90^\circ - \alpha$, with $\tan \beta = \frac{h}{r}$.

TWO DIFFERENT ANGLES

A question asking for “the angle at the vertex” wants 2α ; one asking for “the angle of the slant with the base” wants β . Read which, and mark it on the extracted triangle.

Frustums and inscribed solids

The axial section of a frustum is a trapezium; drop a perpendicular from the top corner to make a right triangle with legs h and $R - r$ and hypotenuse l . Then

$$\tan \beta = \frac{h}{R - r}$$

Inscribed cone in a sphere. The axial section is an isosceles triangle inscribed in a circle. With base radius r and height h in a sphere of radius R :

$$r^2 = h(2R - h) \text{ (the intersecting-chord theorem)}$$

Cylinder in a cone. Similar triangles in the axial section relate the cylinder's radius to its height, and the optimisation of Grade 11 Quarter I then finds the largest one.

Combined problems

A full problem chains three steps:

1. Extract and label the axial section.
2. Solve it with right-angle trigonometry, or the sine or cosine rule.
3. Feed the length found into the volume or surface-area formula.

Example. A cone has slant height 20 and semi-vertical angle 35° . Then $r = 20 \sin 35^\circ \approx 11.47$ and $h = 20 \cos 35^\circ \approx 16.38$, so $V = \frac{1}{3}\pi(11.47)(16.38) \approx 2258$ and $S = \pi r(r + h) \approx 1134$.

02 Worked examples

EXAMPLE 1 A cone has radius 9 and height 12. Find the semi-vertical angle and the angle of the slant with the base.

$$\textcircled{1} \quad \tan \alpha = \frac{9}{12} = 0.75$$

$$\textcircled{2} \quad \alpha \approx 36.9^\circ$$

Semi-vertical.

$$\textcircled{3} \quad \beta = 90 - 36.9 = 53.1^\circ$$

With the base.

ANSWER

$$\alpha \approx 36.9^\circ, \beta \approx 53.1^\circ$$

EXAMPLE 2 A cone has slant height 20 and semi-vertical angle 35° . Find its volume.

$$① \quad r = 20 \sin 35^\circ \approx 11.47$$

$$② \quad h = 20 \cos 35^\circ \approx 16.38$$

$$③ \quad V = \frac{1}{3} \pi (11.47)(16.38)$$

$$④ \quad \approx 2258$$

ANSWER

≈ 2260 cubic units

EXAMPLE 3 A frustum has radii 10 and 4 and height 8. Find the angle the slant makes with the base.

$$① \quad R - r = 6$$

$$② \quad \tan \beta = \frac{8}{6}$$

$$③ \quad \beta \approx 53.1^\circ$$

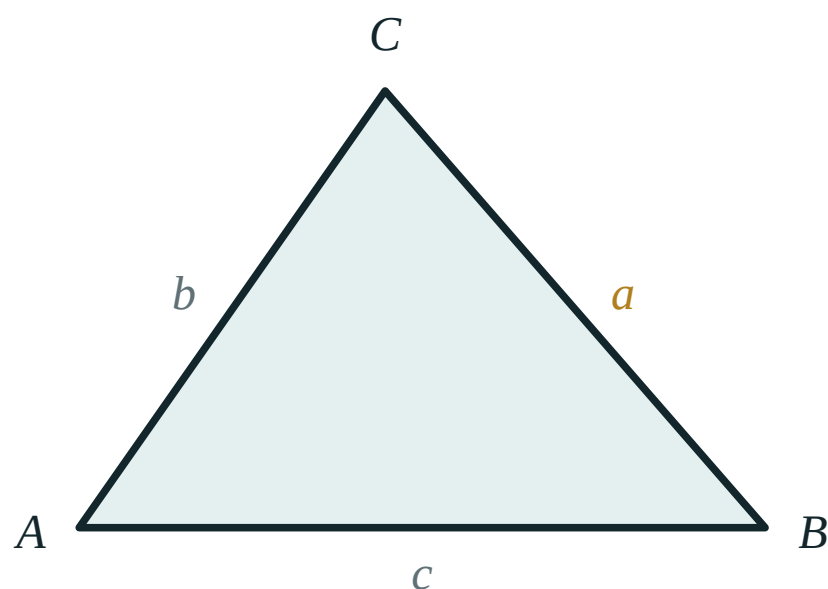
ANSWER

$\approx 53.1^\circ$

03 Interactive model

Cut a cone of plasticine through its axis and hold up the triangle.

The axial triangle


b 7
c 9
A 55°
a 7.6
B 49°
C 76°
area 25.8

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Axial section	O'qli kesim	Осевое сечение
Semi-vertical angle	Yarim uch burchagi	Половина угла при вершине
Slant height	Yasovchi	Образующая
Solid of revolution	Aylanish jismi	Тело вращения
Angle of the slant with the base	Yasovchi va asos burchagi	Угол образующей с основанием
Inscribed cone	Ichki chizilgan konus	Вписанный конус
Extracted triangle	Ajratilgan uchburchak	Выделенный треугольник
Sine rule	Sinuslar teoremasi	Теорема синусов
Cosine rule	Kosinuslar teoremasi	Теорема косинусов

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The axial section of a cone is:

a circle

an isosceles triangle

a rectangle

a trapezium

The axial section of a frustum is:

a triangle

a trapezium

a rectangle

a circle

$\tan \alpha$ for the semi-vertical angle:

$$\frac{h}{r}$$

$$\frac{r}{h}$$

$$\frac{r}{l}$$

$$\frac{h}{l}$$

The angle at the apex is:

$$\alpha$$

$$2\alpha$$

$$90 - \alpha$$

$$180 - \alpha$$

A cone inscribed in a sphere satisfies:

$$r^2 = h(2R - h)$$

$$r = R$$

$$h = R$$

$$r^2 = R^2 + h^2$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Semi-vertical angle with $r = 3, h = 4$

ANSWER $\approx 36.9^\circ$

2. Semi-vertical angle with $r = 5, h = 12$

ANSWER $\approx 22.6^\circ$

3. Angle with the base for $r = 3, h = 4$

ANSWER $\approx 53.1^\circ$

4. r from $l = 10, \alpha = 30^\circ$

ANSWER 5

5. h from $l = 10, \alpha = 30^\circ$

ANSWER $5\sqrt{3}$

6. Axial section of a cylinder

ANSWER a rectangle

7. Axial section of a frustum

ANSWER a trapezium

Medium · the standard question

7 PROBLEMS

1. $r = 9, h = 12$: both angles

ANSWER 36.9° and 53.1°

2. $l = 20, \alpha = 35^\circ$: the volume

ANSWER ≈ 2260

3. Same: the total surface area

ANSWER ≈ 1130

4. Frustum $R = 10, r = 4, h = 8$: the slant angle with the base

ANSWER $\approx 53.1^\circ$

5. Same: the slant height

ANSWER 10

6. A cone with apex angle 60° and $h = 12$: its radius

ANSWER $4\sqrt{3} \approx 6.93$

7. A cone of $r = 6, h = 8$ inscribed in a sphere: the sphere's radius

ANSWER 6.25

Hard · stretch and reason

7 PROBLEMS

1. A cone with semi-vertical angle 30° and volume 1000: its height

ANSWER ≈ 13.1

2. A cylinder of greatest volume inscribed in a cone of $r = 6$, $h = 12$: its dimensions

ANSWER $r = 4$, $h = 4$

3. A cone inscribed in a sphere of radius 10 with height 16: its base radius

ANSWER 8

4. A frustum of slant height 13 and radii 12 and 7: its height and volume

ANSWER 12, ≈ 3630

5. A cone of maximum volume in a sphere of radius R : its height

ANSWER $\frac{4R}{3}$

6. Two cones with the same slant height 15 and angles 20° , 40° : the volume ratio

ANSWER $\approx 1 : 3.5$

7. Prove $r^2 = h(2R - h)$ for a cone in a sphere

ANSWER Intersecting chords through the axis

07 Homework

Homework — 5 tasks

Draw the axial section separately, and mark the angle asked for.

1. A cone has radius 7 and height 24. Find the semi-vertical angle, the apex angle and the angle of the slant with the base.
2. A cone has slant height 26 and semi-vertical angle 40° . Find its radius, height and volume.
3. A frustum has radii 15 and 9 and height 8. Find its slant height and the angle with the base.
4. A cone of base radius 9 and height 12 is inscribed in a sphere. Find the sphere's radius.

5. Explain in three sentences why the axial section reduces a 3-D problem to a 2-D one.

The volume of a sphere and of its parts

Four thirds pi r cubed — where it comes from, and how to measure a cap, a segment and a sector.

LESSON 45–46

2 × 40 MIN

GEOMETRY 11, §20

IGCSE E13.X

LESSON CLOCK

14'

24'

24'

22'

20'

6'

The formula	14 min
Where it comes from	24 min
Caps and hemispheres	24 min
Displacement	22 min
Practice	20 min
Homework	6 min

LEARNING OBJECTIVES

- State and use the volume of a ball.
- Derive the formula by Cavalieri or by integration.
- Compute the volume of a spherical cap and of a hemisphere.
- Solve problems on floating, displacement and compound solids.

TEXTBOOK

National	pp. 223–234
Cambridge	Core & Extended, pp. 355–360
Workbook	Exercise 20.1

01

Explanation

The formula

$$V = \frac{4}{3}\pi R^3$$

R	V	APPROXIMATELY
1	$\frac{4\pi}{3}$	4.19
3	36π	113
6	288π	905
10	$\frac{4000\pi}{3}$	4189

CUBE THE RADIUS, NOT THE DIAMETER

A ball of **diameter** 6 has $R = 3$ and volume 36π , not 288π . Halving first is the commonest error, and it is a factor of eight.

Where the formula comes from

By Cavalieri. Compare a hemisphere of radius R with a cylinder of radius and height R from which a cone has been removed. At height y :

$$\text{hemisphere slice} = \pi(R^2 - y^2); \text{cylinder-minus-cone slice} = \pi R^2 - \pi y^2$$

The slices are equal at every height, so the volumes are equal:

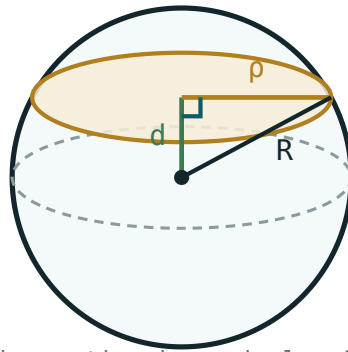
$$V_{\text{hemi}} = \pi R^3 - \frac{1}{3}\pi R^3 = \frac{2}{3}\pi R^3 \Rightarrow V_{\text{ball}} = \frac{4}{3}\pi R^3$$

By integration. Rotating $y = \sqrt{R^2 - x^2}$ about Ox on $[-R, R]$ gives $\pi \int (R^2 - x^2) dx = \frac{4}{3}\pi R^3$

πR^3 — the same answer, from Quarter III algebra.

Caps, hemispheres and sectors

PART	VOLUME
hemisphere	$\frac{2}{3}\pi R^3$
cap of height h	$\frac{\pi h^2}{3}(3R - h)$
spherical sector (cap + cone to the centre)	$\frac{2}{3}\pi R^2 h$
layer between two parallel planes	difference of two caps



the section is a circle of radius ρ
 $\rho^2 = R^2 - d^2$

The cap sits above the cutting plane; its height h is measured from the plane to the top.

Check the cap formula. Put $h = R$: $\frac{\pi R^2}{3}(3R - R) = \frac{2}{3}\pi R^3$ — the hemisphere ✓.

Put $h = 2R$: $\frac{4\pi R^2}{3}(R) = \frac{4}{3}\pi R^3$ — the whole ball ✓.

Displacement

ARCHIMEDES

A solid fully submerged displaces its own volume of liquid. A floating solid displaces its own **weight** of liquid.

Example. A ball of radius 3 cm is dropped into a cylinder of radius 5 cm containing water. By how much does the level rise?

$$36\pi = \pi(25)\Delta h \Rightarrow \Delta h = \frac{36}{25} = 1.44 \text{ cm}$$

This is how an irregular solid's volume is measured, and it is the practical purpose of the formula.

02 Worked examples

EXAMPLE 1 Find the volume of a ball of diameter 12 cm.

① $R = 6$
Halve the diameter first.

② $V = \frac{4}{3}\pi(216)$

③ $= 288\pi \approx 905 \text{ cm}^3.$

ANSWER

$$288\pi \approx 905 \text{ cm}^3$$

EXAMPLE 2 A cap of height 4 is cut from a sphere of radius 10. Find its volume.

① $V = \frac{\pi(16)}{3}(30 - 4)$

② $= \frac{16\pi}{3} \times 26$

③ $= \frac{416\pi}{3} \approx 436$

ANSWER

$$\frac{416\pi}{3} \approx 436$$

EXAMPLE 3 A ball of radius 3 is dropped into a cylinder of radius 5. Find the rise in the water level.

① Ball volume 36π .

② $\pi(25)\Delta h = 36\pi$

③ $\Delta h = 1.44$ cm.

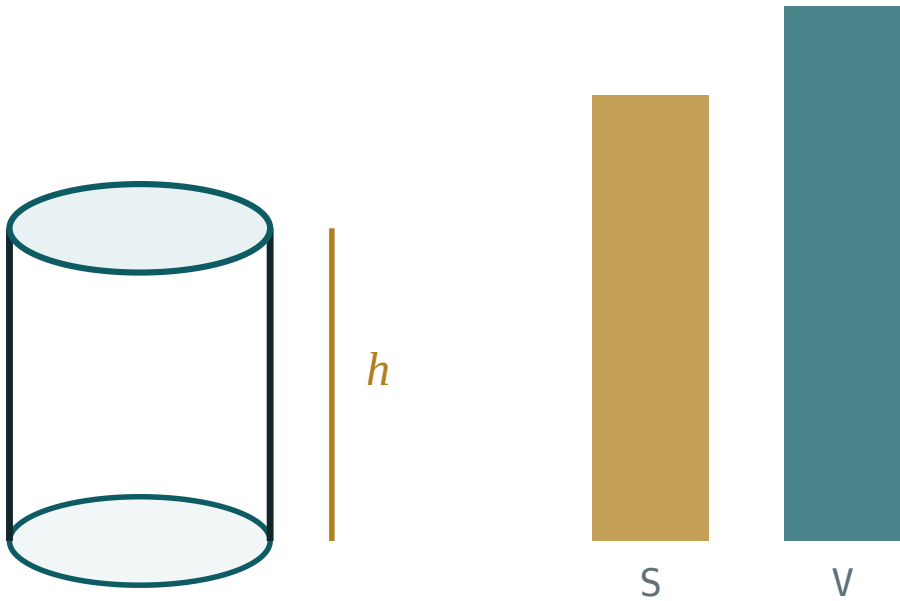
ANSWER

1.44 cm

03 Interactive model

Drop a ball into a measuring cylinder and read the rise before computing it.

The sphere against its cylinder



base area B **50.3**

perimeter P **25.1**

total S **251.3**

volume V **301.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Ball	Shar	Шар
Hemisphere	Yarim shar	Полушар
Spherical cap	Sharsimon segment	Шаровой сегмент
Spherical sector	Sharsimon sektor	Шаровой сектор
Spherical layer	Sharsimon qatlam	Шаровой слой
Displacement	Siqib chiqarish	Вытеснение
Cavalieri's principle	Kavaleri prinsipi	Принцип Кавальери
Height of a cap	Segment balandligi	Высота сегмента
Archimedes' theorem	Arximed teoremasi	Теорема Архимеда

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of a ball is:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$\frac{2}{3}\pi R^3$$

$$\pi R^3$$

A ball of diameter 6 has volume:

$$36\pi$$

$$288\pi$$

$$72\pi$$

$$108\pi$$

A hemisphere of radius R :

$$\frac{4}{3}\pi R^3$$

$$\frac{2}{3}\pi R^3$$

$$\frac{1}{3}\pi R^3$$

$$2\pi R^3$$

A sphere is what fraction of its enclosing cylinder?

$$\frac{1}{2}$$

$$\frac{2}{3}$$

$$\frac{3}{4}$$

$$\frac{1}{3}$$

A submerged solid displaces:

its weight

its volume

half its volume

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume, $R = 3$

ANSWER 36π

2. Volume, $R = 6$

ANSWER 288π

3. Volume, $R = 1$

ANSWER $\frac{4\pi}{3}$

4. Volume of a ball of diameter 10

ANSWER $\frac{500\pi}{3}$

5. Hemisphere, $R = 3$

ANSWER 18π

6. A ball is what fraction of its cylinder?

ANSWER $\frac{2}{3}$

7. Volume, $R = 10$

ANSWER $\frac{4000\pi}{3}$

Medium · the standard question

7 PROBLEMS

1. Ball of diameter 12: its volume

ANSWER 288π

2. Cap of height 4 in a sphere of radius 10

ANSWER $\frac{416\pi}{3}$

3. Ball $R = 3$ in a cylinder $r = 5$: the rise

ANSWER 1.44 cm

4. A ball of volume 972π : its radius

ANSWER 9

5. A hemisphere of radius 6: volume and curved area

ANSWER $144\pi, 72\pi$

6. A cap of height 2 in a sphere of radius 5

ANSWER $\frac{52\pi}{3}$

7. Two balls of radii 3 and 6: the volume ratio

ANSWER $1 : 8$

Hard · stretch and reason

7 PROBLEMS

1. A sphere of radius 10 cut by planes at 4 and 7 from the centre: the layer volume

ANSWER ≈ 314

2. A ball just fits in a cube of edge 12: the unused volume

ANSWER $1728 - 288\pi \approx 823$

3. Derive $V = \frac{4}{3}\pi R^3$ by integration

ANSWER $\pi \int (R^2 - x^2) dx$ over $[-R, R]$

4. A spherical cap of volume $\frac{16\pi}{3}$ in a sphere of radius 4: its height

ANSWER 2

5. A ball of radius r floats with half submerged. Its density relative to water

ANSWER 0.5

6. A hemisphere of radius 6 filled with water is poured into a cone $r = 6$. The depth

ANSWER 12

7. Three balls of radius 2 in a cylinder $r = 2, h = 12$: the empty fraction

ANSWER $\frac{1}{3}$

07 Homework

Homework — 5 tasks

Halve the diameter before cubing, every time.

- Find the volume of a ball of radius 7 cm and of one of diameter 9 cm.
- A cap of height 3 is cut from a sphere of radius 12. Find its volume.
- A ball of radius 4 is dropped into a cylinder of radius 6. Find the rise in the level.

4. A ball has volume $\frac{2048\pi}{3}$. Find its radius and its surface area.
5. Explain in three sentences why a sphere is exactly two thirds of its enclosing cylinder.

The surface area of a sphere

Four times the area of a great circle — the second half of Archimedes’ result.

LESSON 47

1 × 40 MIN

GEOMETRY 11, §21

IGCSE E13.X

LESSON CLOCK

10'

12'

12'

8'

3'

The formula	10 min
Archimedes' theorem	12 min
Hemispheres and caps	12 min
Combined problems	8 min
Homework	3 min

LEARNING OBJECTIVES

- State and use $S = 4\pi R^2$.
- Relate the sphere’s area to that of its enclosing cylinder.
- Compute the curved area of a hemisphere and of a cap.
- Solve problems combining area and volume.

TEXTBOOK

National	pp. 235–240
Cambridge	Core & Extended, pp. 361–364
Workbook	Exercise 21.1

01

Explanation

The formula

$S = 4\pi R^2$

Four times the area of a great circle. A useful mental check: unroll a sphere and it covers exactly four of its own silhouettes.

R	S	V
1	4π	$\frac{4\pi}{3}$
3	36π	36π
6	144π	288π

At $R = 3$ the two numbers coincide — a coincidence of units, not a theorem. It happens because $4\pi R^2 = \frac{4}{3}\pi R^3$ exactly when $R = 3$.

Archimedes’ theorem

THE SPHERE AND ITS CYLINDER

For a sphere inscribed in a cylinder of radius R and height $2R$:

| sphere : cylinder = 2 : 3 for **both** volume and total surface area

Volumes: $\frac{4}{3}\pi R^3$ against $2\pi R^3$. Areas: $4\pi R^2$ against $2\pi R^2 + 4\pi R^2 = 6\pi R^2$. Both give $\frac{2}{3}$.

More sharply still: the sphere's surface area equals the cylinder's **curved** area exactly. Any horizontal band of the sphere has the same area as the corresponding band of the cylinder — the projection stretches horizontally exactly as much as it squashes vertically. That is the theorem Archimedes asked to have carved on his tomb.

Hemispheres and caps

PART	CURVED AREA	TOTAL AREA
sphere	$4\pi R^2$	$4\pi R^2$
hemisphere	$2\pi R^2$	$3\pi R^2$
cap of height h	$2\pi Rh$	$2\pi Rh + \pi r^2$
zone between two planes h apart	$2\pi Rh$	—

A SOLID HEMISPHERE HAS A FLAT FACE

Its **curved** area is $2\pi R^2$, but a solid hemisphere also has the flat circle πR^2 , giving $3\pi R^2$. Read whether the object is a bowl or a solid dome.

The zone result is remarkable. The area of a band depends only on its height h , not on where it sits. A band near the equator and one near the pole, of equal height, have equal area.

Combined problems

Given one of S or V , the other follows through R :

$$R = \sqrt{\frac{S}{4\pi}} \quad R = \sqrt[3]{\frac{3V}{4\pi}}$$

Example. A ball has surface area 100π . Then $R = 5$ and $V = \frac{500\pi}{3} \approx 524$.

Useful relation. $\frac{V}{S} = \frac{R}{3}$ for every sphere — so a bigger ball has proportionally less surface per unit volume, which is the $\frac{1}{k}$ rule of Lesson 40 again.

02 Worked examples

EXAMPLE 1 Find the surface area of a ball of radius 7 cm.

$$1 \quad S = 4\pi(49)$$

$$2 \quad = 196\pi$$

$$3 \quad \approx 616 \text{ cm}^2.$$

ANSWER

$$196\pi \approx 616 \text{ cm}^2$$

EXAMPLE 2 A ball has surface area 144π . Find its radius and volume.

$$1 \quad 4\pi R^2 = 144\pi \Rightarrow R^2 = 36$$

$$2 \quad R = 6$$

$$3 \quad V = \frac{4}{3}\pi(216) = 288\pi \approx 905$$

ANSWER

$$R = 6, V = 288\pi$$

EXAMPLE 3 A solid hemisphere has radius 5. Find its total surface area.

$$1 \quad \text{Curved } 2\pi(25) = 50\pi.$$

$$2 \quad \text{Flat face } \pi(25) = 25\pi.$$

$$3 \quad 75\pi \approx 236.$$

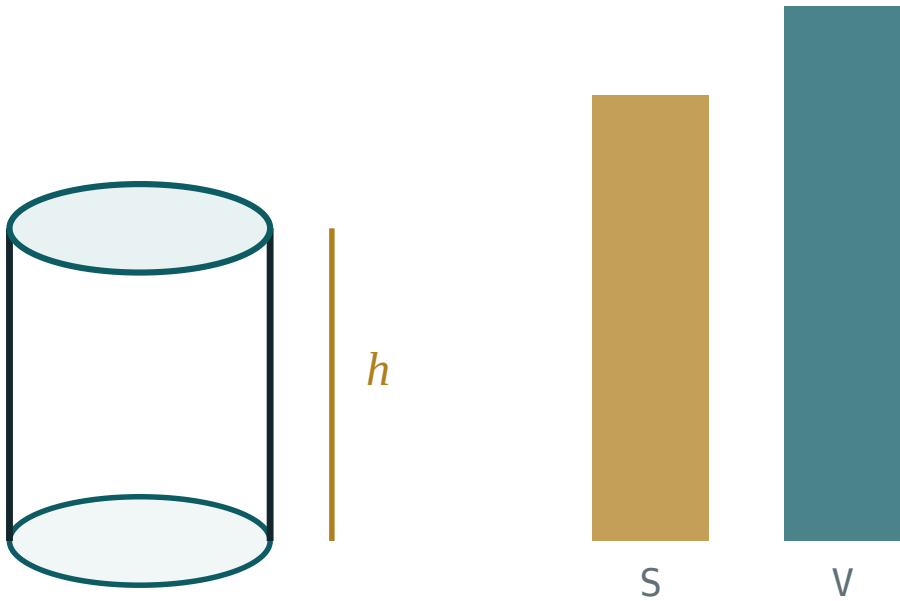
ANSWER

$$75\pi \approx 236$$

03 Interactive model

Wrap string round a ball and compare with four traced great circles.

Area against volume



base area B **50.3**

perimeter P **25.1**

total S **251.3**

volume V **301.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Surface area of a sphere	Sfera sirti	Площадь сферы
Great circle	Katta aylana	Большая окружность
Hemisphere	Yarim shar	Полушар
Curved surface	Yon sirt	Боковая поверхность
Zone (spherical belt)	Sharsimon belbog'	Шаровой пояс
Archimedes' theorem	Arximed teoremasi	Теорема Архимеда
Enclosing cylinder	Tashqi silindr	Описанный цилиндр
Cap	Segment	Сегмент

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The surface area of a sphere is:

πR^2

$2\pi R^2$

$4\pi R^2$

$\frac{4}{3}\pi R^3$

A hemisphere's curved area:

πR^2

$2\pi R^2$

$3\pi R^2$

$4\pi R^2$

A solid hemisphere's total area:

$2\pi R^2$

$3\pi R^2$

$4\pi R^2$

πR^2

A cap of height h has curved area:

$$\pi Rh$$

$$2\pi Rh$$

$$4\pi Rh$$

$$\pi h^2$$

Sphere : cylinder for area is:

$$1 : 2$$

$$2 : 3$$

$$1 : 3$$

$$3 : 4$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Area, $R = 3$

ANSWER 36π

2. Area, $R = 5$

ANSWER 100π

3. Area, $R = 7$

ANSWER 196π

4. Area of a ball of diameter 10

ANSWER 100π

5. Hemisphere curved area, $R = 4$

ANSWER 32π

6. Solid hemisphere total area, $R = 4$

ANSWER 48π

7. Cap curved area, $R = 10, h = 3$

ANSWER 60π

Medium · the standard question

7 PROBLEMS

1. Area 144π : the radius and volume

ANSWER $6, 288\pi$

2. Solid hemisphere $R = 5$: total area

ANSWER 75π

3. A ball of volume 972π : its surface area

ANSWER 324π

4. A zone of height 4 on a sphere of radius 9

ANSWER 72π

5. Sphere : cylinder area ratio

ANSWER $2 : 3$

6. $\frac{V}{S}$ for a sphere of radius 12

ANSWER 4

7. Two balls of radii 2 and 6: the area ratio

ANSWER $1 : 9$

Hard · stretch and reason

7 PROBLEMS

1. A ball of area S : its volume in terms of S

ANSWER $\frac{S}{3} \sqrt{\frac{S}{4\pi}}$

2. A hemispherical bowl of inner radius 9 and outer 10: the material volume

ANSWER $\frac{2\pi}{3} (1000 - 729) \approx 568$

3. A sphere and a cube have equal surface areas. Which has the greater volume?

ANSWER The sphere

4. A cap of curved area 60π on a sphere of radius 10: its height

ANSWER 3

5. Prove the zone area depends only on h

ANSWER Archimedes' projection is area-preserving

6. A ball is painted at 2 so'm per cm^2 . Cost for $R = 20$

ANSWER $3200\pi \approx 10\,050$ so'm

7. A sphere of radius R inside a cube of edge $2R$: the area ratio

ANSWER $\pi : 6$

07 Homework

Homework — 4 tasks

Say whether the object is hollow or solid before choosing the formula.

- Find the surface area of a ball of radius 11 cm and of one of diameter 14 cm.
- A ball has surface area $324\pi \text{ cm}^2$. Find its radius and volume.
- Find the total surface area of a solid hemisphere of radius 8 cm.
- State Archimedes' theorem and verify both ratios for $R = 5$.

Compound solids and real measurement [Cambridge insert]

Cut a real object into named solids, add or subtract, and carry the bounds through.

LESSON 48–50

3 × 40 MIN

GEOMETRY 11, AMALIY MASHG‘ULOT

IGCSE E13.X COMPOUND SOLIDS · E13.3 BOUNDS

LESSON CLOCK

12

25'

25'

30'

30'

13'

Reading an object as solids	12 min
Adding and subtracting volume	25 min
Which faces survive	25 min
Bounds through a compound	30 min
Measuring the classroom	30 min
Homework	13 min

LEARNING OBJECTIVES

- Decompose a real object into prisms, cylinders, cones, pyramids and spheres.
- Add and subtract volumes, and decide which faces survive into the surface area.
- Carry upper and lower bounds through a compound calculation.
- Report a measured answer to a sensible degree of accuracy.

TEXTBOOK

National	pp. 241–248
Cambridge	Core & Extended, pp. 365–374
Workbook	Exercise 13.5

01

Explanation

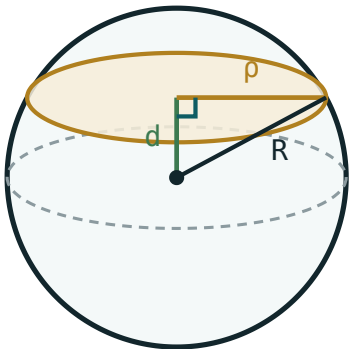
Reading an object as solids

Nothing new is needed for the next three lessons — only the seven formulas already proved. The whole skill is **reading**: seeing a grain silo as a cylinder with a hemisphere on top, a pencil as a cylinder with a cone, a water tower as a sphere on a thin cylinder.

OBJECT	READ AS	VOLUME
grain silo	cylinder + hemisphere	$\pi r^2 h + \frac{2}{3} \pi r^3$
pencil	cylinder + cone	$\pi r^2 h + \frac{1}{3} \pi r^2 k$
ice-cream	cone + hemisphere	$\frac{1}{3} \pi r^2 h + \frac{2}{3} \pi r^3$
a bead	sphere – cylinder	$\frac{4}{3} \pi R^3 - \pi r^2 \cdot 2R$
a pipe	cylinder – cylinder	$\pi(R^2 - r^2)L$
a mosque dome	hemisphere – hemisphere	$\frac{2}{3} \pi(R^3 - r^3)$

THE ONE RULE FOR THE CUT

Cut along a plane where two named solids meet, and the cut face is always a **circle** or a **polygon** shared by both. If your cut leaves a face that is neither, you have cut in the wrong place.



the section is a circle of radius ρ
 $\rho^2 = R^2 - d^2$

*A plane cut through a solid of revolution leaves a circle
— the shared face of the two parts.*

Adding and subtracting volume

Volume is easy: add the parts, or subtract the hollow. Nothing is shared, nothing is double-counted, because two solids that meet along a plane have no common interior.

$$V_{compound} = V_1 + V_2 + \dots \text{ or } V_{solid} - V_{hole}$$

Worked case — the silo. Cylinder $r = 4$ m, $h = 12$ m, with a hemispherical cap of the same radius:

PART	FORMULA	VALUE
cylinder	$\pi r^2 h$	192π
hemisphere	$\frac{2}{3}\pi r^3$	$\frac{128\pi}{3}$
total	—	$\frac{704\pi}{3} \approx 737\text{ m}^3$

KEEP π TO THE LAST LINE

$192\pi + \frac{128\pi}{3}$ is exact; $603.2 + 134.0$ is not. Round once, at the end, and never feed a rounded number into the next part.

Which faces survive

Surface area is where the marks go. When two solids are joined, the two faces that meet **disappear** — they are now inside. The surface area of the compound is *not* the sum of the two surface areas.

$$S_{compound} = S_1 + S_2 - 2 \times (\text{shared face})$$

The silo again. The cylinder's top circle and the hemisphere's flat face are the same circle, and both are now internal:

FACE	AREA	COUNTS?
cylinder base	$\pi(16) = 16\pi$	yes — it stands on the ground
cylinder curved	$2\pi(4)(12) = 96\pi$	yes
cylinder top	16π	no — covered
hemisphere flat	16π	no — covered
hemisphere curved	$2\pi(16) = 32\pi$	yes

So $S = 16\pi + 96\pi + 32\pi = 144\pi \approx 452 \text{ m}^2$. If the silo is painted only on the outside above the ground, drop the base too: $128\pi \approx 402 \text{ m}^2$.

THREE QUESTIONS BEFORE WRITING A NUMBER

- Is the object **solid or hollow**? A bowl has an inside surface as well.
- Does it **stand on** something? Then its base may not be painted, wetted or covered.
- Which faces are **joined**? Those never count.

Bounds through a compound

Every measurement is a range. A silo measured as $r = 4 \text{ m}$ and $h = 12 \text{ m}$ to the nearest metre means $3.5 \leq r < 4.5$ and $11.5 \leq h < 12.5$.

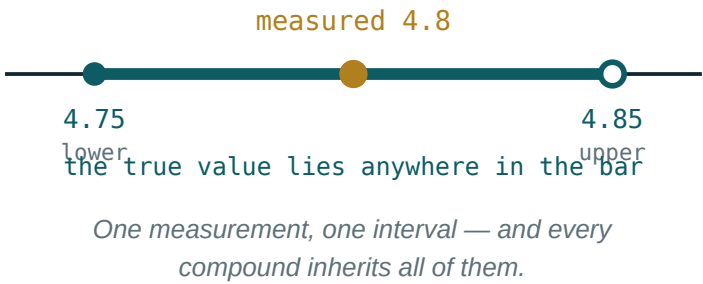
Volume grows with both, so the extremes pair up directly:

	R	H	V
lower	3.5	11.5	$\approx 532 \text{ m}^3$
stated	4	12	$\approx 737 \text{ m}^3$
upper	4.5	12.5	$\approx 986 \text{ m}^3$

A SUBTRACTION REVERSES ONE BOUND

For a pipe, $V = \pi(R^2 - r^2)L$. The **largest** metal volume needs the largest R and the **smallest** r . Whenever a quantity is subtracted, swap its two bounds.

Reporting. The silo's volume lies between 532 and 986 m³ — the two bounds do not agree even to one significant figure, so 700 m³ is the honest answer, not 737.4. Report to the accuracy the bounds support.



Measuring the classroom

Lesson 50 is practical. In pairs, choose one object in the school, measure it, and produce a single A4 sheet:

1. a sketch, decomposed into named solids, with every measurement marked;
2. the instrument used and its precision (“tape, nearest 0.5 cm”);
3. the volume, exact then rounded;
4. the surface area, saying which faces you excluded and why;
5. upper and lower bounds, and the accuracy you can honestly claim.

OBJECT	DECOMPOSITION	WATCH FOR
a waste bin	frustum of a cone	open top — no lid area
a football	sphere	measure the circumference, not the radius
a chalk stick	cylinder, or a frustum	the taper is real
the door frame	three cuboids	the corners are shared
a water bottle	cylinder + frustum + cylinder	capacity ≠ outer volume
a lamp shade	frustum, curved surface only	two circles missing

CAPACITY IS NOT VOLUME

A bottle's **capacity** uses inner dimensions; its **volume** as a solid object uses outer ones. The difference is the glass, and in an exam the word chosen tells you which one is wanted.

02 Worked examples

EXAMPLE 1 An ice-cream is a cone of radius 3 cm and height 10 cm topped by a hemisphere of the same radius. Find its volume.

1 Cone $\frac{1}{3}\pi(9)(10) = 30\pi$.

2 Hemisphere $\frac{2}{3}\pi(27) = 18\pi$.

3 $V = 48\pi$
Add — nothing overlaps.

4 $\approx 151 \text{ cm}^3$.

ANSWER

$48\pi \approx 151 \text{ cm}^3$

EXAMPLE 2 Find the surface area of the same ice-cream (the cone is open at the top, under the hemisphere).

1 Slant $l = \sqrt{9 + 100} = \sqrt{109} \approx 10.44$.

2 Cone curved $\pi rl = 3\pi\sqrt{109} \approx 98.4$.
No base circle — it is joined.

3 Hemisphere curved $2\pi(9) = 18\pi \approx 56.5$.
Its flat face is joined too.

4 $S \approx 155 \text{ cm}^2$.

ANSWER

$\approx 155 \text{ cm}^2$

EXAMPLE 3 A wooden bead is a sphere of radius 5 mm drilled through the centre by a hole of radius 2 mm. Find the volume of wood.

① Sphere $\frac{4}{3}\pi(125) = \frac{500\pi}{3}$.

② The hole is a cylinder of length 10 mm: $\pi(4)(10) = 40\pi$.
An approximation — the true hole has spherical ends.

③ $V \approx \frac{500\pi}{3} - 40\pi = \frac{380\pi}{3}$

④ $\approx 398 \text{ mm}^3$.

ANSWER

$$\frac{380\pi}{3} \approx 398 \text{ mm}^3$$

EXAMPLE 4 A cylindrical tank has radius 2.0 m and height 5.0 m, both to the nearest 0.1 m. Find the bounds of its capacity in litres.

① $1.95 \leq r < 2.05, 4.95 \leq h < 5.05$.

② $V_{\min} = \pi(1.95^2)(4.95) \approx 59.13 \text{ m}^3$.

③ $V_{\max} = \pi(2.05^2)(5.05) \approx 66.68 \text{ m}^3$.

④ Between 59 100 and 66 700 litres.
So “about 60 000 litres”.

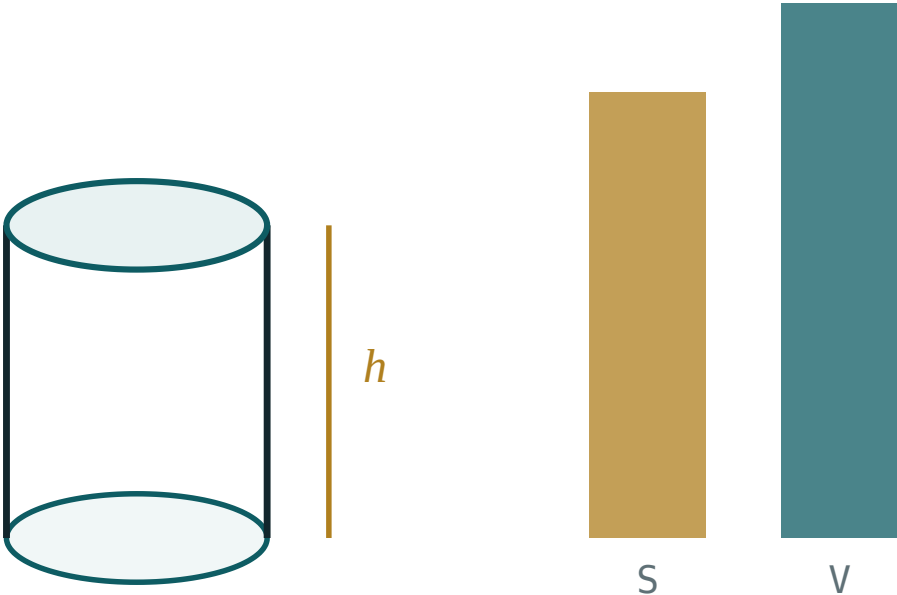
ANSWER

$\approx 59\,100$ to $66\,700$ litres

03 Interactive model

Bring a real bottle, a real ball and a tape measure. Measure in front of the class, then argue about the last digit.

A cylinder, and what a cap adds



The diagram shows a 3D cylinder with a light blue top and bottom circular face and a dark teal side. To its right is a vertical orange line segment labeled h . Further right are two vertical bars: a shorter orange bar labeled S and a taller teal bar labeled V .

base area B 50.3

perimeter P 25.1

total S 251.3

volume V 301.6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Compound solid	Murakkab jism	Составное тело
Decomposition	Qismlarga ajratish	Разбиение
Hollowed solid	Bo'shliqli jism	Полое тело
Capacity	Sig'im	Вместимость
Density	Zichlik	Плотность
Upper bound	Yuqori chegara	Верхняя граница
Lower bound	Quyi chegara	Нижняя граница
Degree of accuracy	Aniqlik darajasi	Степень точности
Wetted surface	Ho'llanadigan sirt	Смачиваемая поверхность
Silo	Silos minorasi	Силосная башня

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

When two solids are joined, the shared face:

counts twice

counts once

does not count at all

is halved

Volumes of joined solids are:

added

multiplied

averaged

subtracted

For the largest metal volume in a pipe take:

both bounds large

largest R , smallest r

both small

the stated values

A silo is a cylinder plus:

a cone

a hemisphere

a prism

a frustum

Capacity uses:

outer dimensions

inner dimensions

either

the mean

Rounding should happen:

at each step

only at the end

never

twice

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Cylinder $r = 2$, $h = 5$ plus hemisphere $r = 2$: volume

ANSWER $20\pi + \frac{16\pi}{3} \approx 79.6$

2. Cone $r = 3$, $h = 4$ plus hemisphere $r = 3$: volume

ANSWER $12\pi + 18\pi = 30\pi$

3. Pipe $R = 5$, $r = 3$, $L = 20$: metal volume

ANSWER 320π

4. Cube edge 6 with a cylindrical hole $r = 1$ through it: volume

ANSWER $216 - 6\pi \approx 197$

5. Cylinder $r = 3$, $h = 10$ with a cone cap $h = 4$: volume

ANSWER $90\pi + 12\pi = 102\pi$

6. Bounds of 8.0 cm to the nearest 0.1 cm

ANSWER 7.95 and 8.05

7. Surface area of a solid hemisphere on a cylinder base $r = 2$, $h = 3$ (closed bottom)

ANSWER $4\pi + 12\pi + 8\pi = 24\pi$

Medium · the standard question

7 PROBLEMS

1. Silo: cylinder $r = 4$, $h = 12$ plus hemisphere. Volume

ANSWER $\frac{704\pi}{3} \approx 737 \text{ m}^3$

2. Same silo: painted outside, base excluded

ANSWER $128\pi \approx 402 \text{ m}^2$

3. Ice-cream: cone $r = 3$, $h = 10$ plus hemisphere. Surface area

ANSWER $\approx 155 \text{ cm}^2$

4. A bead: sphere $R = 5$ drilled by $r = 2$ (straight hole). Wood volume

ANSWER $\frac{380\pi}{3} \approx 398$

5. Tank $r = 2.0$, $h = 5.0$ m to 0.1 m: capacity bounds in litres

ANSWER $\approx 59\,100$ to $66\,700$

6. A dome: hemisphere $R = 6$ hollowed to $r = 5$. Material volume

ANSWER $\frac{2\pi}{3}(216 - 125) \approx 191$

7. A pencil: cylinder $r = 0.4$, $h = 15$ plus cone $h = 2$ cm. Volume

ANSWER $2.4\pi + 0.107\pi \approx 7.87 \text{ cm}^3$

Hard · stretch and reason

7 PROBLEMS

1. A silo of the same shape must hold 1000 m^3 with $r = 4$. Find h

ANSWER $h \approx 17.2 \text{ m}$

2. Pipe $R = 9$, $r = 7 \text{ cm}$, $L = 200 \text{ cm}$, all to the nearest cm: bounds of the metal volume

ANSWER $\approx 18\,200$ to $\approx 22\,200 \text{ cm}^3$

3. A cone of radius r is capped by a hemisphere of radius r . For what h are the two volumes equal?

ANSWER $h = 2r$

4. An open lamp shade is a frustum, radii 10 and 20 , slant 26 . Find the material area

ANSWER $780\pi \approx 2450 \text{ cm}^2$

5. A bottle: cylinder $r = 3$, $h = 12$, frustum $3 \rightarrow 1.5$, $h = 4$, neck $r = 1.5$, $h = 3$. Capacity in ml

ANSWER $\approx 421 \text{ ml}$

6. A hemispherical bowl of inner radius 9 cm is filled to a depth of 4 cm . Find the volume of water

ANSWER $\frac{\pi}{3}(4^2)(27 - 4) \approx 385 \text{ cm}^3$

7. A solid is a cube of edge $2R$ with the inscribed sphere removed. Find the fraction of the cube remaining

ANSWER $1 - \frac{\pi}{6} \approx 0.476$

07 Homework

Homework — 5 tasks

Task 5 is the measured object; bring the A4 sheet and the object itself if you can.

1. A grain silo is a cylinder of radius 3 m and height 10 m with a hemispherical top. Find its volume and its outside surface area, excluding the base.

2. A pipe has outer radius 8 cm, inner radius 6 cm and length 1.5 m. Find the volume of metal and its mass at 7.8 g/cm³.
3. A tank measures $r = 1.5$ m and $h = 4.0$ m, each to the nearest 0.1 m. Find the bounds of its capacity in litres, and state the answer to a sensible accuracy.
4. An ice-cream cone of radius 3.5 cm and height 12 cm carries a hemisphere of the same radius. Find the total volume, and explain which two faces you did not include in the surface area.
5. Measure one object in your home. Produce the A4 sheet: sketch, decomposition, instrument and precision, volume, surface area, bounds.

Control work 3, and work on the mistakes

Pyramids, cones and spheres in one paper, then the quarter drawn as a single map.

LESSON 51–52

2 × 40 MIN

GEOMETRY 11, NAZORAT ISHI 3

IGCSE E13 REVIEW

LESSON CLOCK

3

42'

12'

20'

10'

3

Instructions	3 min
The paper	42 min
Answers	12 min
Rewrite	20 min
The map	10 min
Targets	3 min

LEARNING OBJECTIVES

- Apply the formulas of Quarter III in one assessment.
- Choose correctly between slant height, height and lateral edge.
- Classify each lost mark and rewrite the solution in full.
- Draw the quarter as one map, from the pyramid to the sphere.

TEXTBOOK

National	pp. 249–252
Cambridge	Core & Extended, pp. 375–378
Workbook	Control paper G3

01

Explanation

The paper — 35 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Regular square pyramid, base edge 10, height 12: apothem, lateral area, total area	6	L33–34
2	The same pyramid: its volume, and the angle its lateral edge makes with the base	6	L33–34, L43–44
3	A cone of radius 5 and slant 13: height, curved area, volume, and the sector angle of its net	7	L36–38
4	A frustum of a cone, radii 6 and 3, height 4: volume and curved area	6	L39
5	A ball of surface area 144π : its radius and volume; then the volume of a cap of height 2	5	L41–42, L45–47
6	Two similar cones have volumes 27 and 125: the ratios of heights and of surface areas	5	L40

WHERE THE MARKS ARE ACTUALLY LOST

Q1 and Q3 each carry a mark for using the **slant height** in an area and the **vertical height** in a volume. Q4 carries one for not treating the frustum as a prism. Q5 carries one for reading “surface area” rather than “volume”. Four of the thirty-five marks are for reading.

Mark by mark

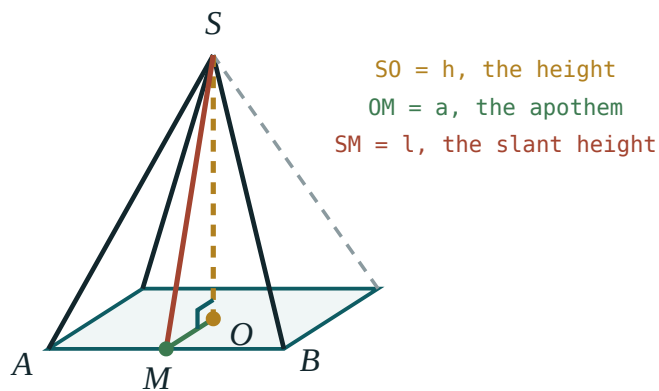
SLIP	WHAT IT LOOKS LIKE	THE FIX
slant for height	$V = \frac{1}{3}Bl$	volume always uses h , the perpendicular
height for slant	$S = \frac{1}{2}Ph$	area always uses m , the apothem
frustum by subtraction only	finding the whole cone first, then failing	use $\frac{\pi h}{3}(R^2 + Rr + r^2)$ directly
the missing $\frac{1}{3}$	$V = Bh$	pyramids and cones are a third of the prism
sphere formulas swapped	$S = \frac{4}{3}\pi R^3$	area has R^2 , volume has R^3
ratios not cubed	$V_1 : V_2 = k$	k, k^2, k^3

For every mark lost, write the slip's name in the margin and then rewrite the whole question — not the line that went wrong. A rewritten line is forgotten by Friday; a rewritten solution is not.

The quarter as one map

Six boxes and the links as sentences:

- **prism** → **pyramid** — “a third of it, on the same base and height”
- **cylinder** → **cone** — “the same third, with $B = \pi r^2$ ”
- **cone** → **frustum** — “cut off the top, or use $R^2 + Rr + r^2$ ”
- **net** → **curved area** — “the cone unrolls into a sector, $S = \pi r l$ ”
- **sphere** → **its cylinder** — “2 : 3 for both volume and area”
- **scale k** → **k, k^2, k^3** — “length, area, volume”



$$l^2 = h^2 + a^2 \quad S_{\text{lat}} = \frac{1}{2}Pl$$

The three lengths of the first lesson — still the whole of Q1 and Q2.

SEVEN FORMULAS, ONE PAGE

$$V_{\text{pyr}} = \frac{1}{3}Bh \quad S_{\text{lat}} = \frac{1}{2}Pl \quad V_{\text{cone}} = \frac{1}{3}\pi r^2 h \quad S_{\text{cone}} = \pi rl$$

$$V_{\text{frus}} = \frac{\pi h}{3}(R^2 + Rr + r^2) \quad V_{\text{sph}} = \frac{4}{3}\pi R^3 \quad S_{\text{sph}} = 4\pi R^2$$

If these seven sit on one page in your own handwriting, Quarter IV needs nothing new.

Looking forward

Quarter IV adds no new solid. It is application: containers, costs, optimisation and the entrance paper, where a question rarely says which formula it wants. The reading skill of Lesson 48–50 becomes the whole of the work.

ONE HABIT TO CARRY FORWARD

Before any arithmetic, write down the object, its decomposition and the letter of every length you know. Almost every mark lost this quarter was lost before the first calculation.

02 Worked examples

EXAMPLE 1 Model answer, Q1: a regular square pyramid, base edge 10, height 12.

① Apothem of the base 5; so $m = \sqrt{144 + 25} = 13$.

The first right triangle.

② $S_{lat} = \frac{1}{2}(40)(13) = 260$.

Perimeter times apothem, halved.

③ $B = 100$.

④ $S_{total} = 360$.

ANSWER

$$m = 13, S_{lat} = 260, S_{total} = 360$$

EXAMPLE 2 Model answer, Q3: a cone of radius 5 and slant 13.

$$① \quad h = \sqrt{169 - 25} = 12.$$

$$② \quad S_{\text{curved}} = \pi(5)(13) = 65\pi.$$

Slant, not height.

$$③ \quad V = \frac{1}{3}\pi(25)(12) = 100\pi.$$

Height, not slant.

$$④ \quad \text{Sector angle } \theta = 360^\circ \times \frac{5}{13} \approx 138.5^\circ.$$

$$\theta = 360^\circ \frac{r}{l}.$$

ANSWER

$$h = 12, 65\pi, 100\pi, \approx 138.5^\circ$$

EXAMPLE 3 Model answer, Q4: a frustum, radii 6 and 3, height 4.

$$① \quad V = \frac{\pi(4)}{3}(36 + 18 + 9) = 84\pi.$$

Use the formula directly.

$$② \quad \text{Slant } l = \sqrt{16 + 9} = 5.$$

From the difference of radii.

$$③ \quad S_{\text{curved}} = \pi(6 + 3)(5) = 45\pi.$$

$$④ \quad V \approx 264, S \approx 141.$$

ANSWER

$$84\pi \text{ and } 45\pi$$

EXAMPLE 4 Model answer, Q6: two similar cones of volumes 27 and 125.

$$① \quad k^3 = \frac{125}{27}$$

$$② \quad k = \frac{5}{3}$$

Heights are in 3 : 5.

$$③ \quad k^2 = \frac{25}{9}$$

④ Surface areas are in 9 : 25.

ANSWER

3 : 5 and 9 : 25

03 Interactive model

Work Q3 on the board twice — once correctly, once with slant and height swapped — and let the class find the difference.

The quarter in ten questions

Volume of a pyramid:

Bh

$\frac{1}{3} Bh$

$\frac{1}{2} Bh$

$3Bh$

Lateral area of a regular pyramid:

$\frac{1}{2} Ph$

$\frac{1}{2} Pm$

Pm

Bm

Curved area of a cone:

πr^2

$\pi r l$

$\pi r h$

$2\pi r l$

Volume of a cone:

$$\pi r^2 h$$

$$\frac{1}{3} \pi r^2 h$$

$$\frac{1}{3} \pi r^2 l$$

$$\pi r l$$

The sector angle of a cone's net:

$$360^\circ \frac{r}{l}$$

$$360^\circ \frac{l}{r}$$

$$180^\circ \frac{r}{l}$$

$$90^\circ$$

Volume of a frustum of a cone:

$$\frac{\pi h}{3} (R^2 + r^2)$$

$$\frac{\pi h}{3} (R^2 + Rr + r^2)$$

$$\pi h (R^2 - r^2)$$

$$\frac{\pi h}{2} (R + r)$$

Volume of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$\frac{2}{3}\pi R^3$$

$$\pi R^3$$

Surface area of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$2\pi R^2$$

$$\pi R^2$$

A plane section of a sphere is:

an ellipse

a circle

a parabola

a point

Scale factor k multiplies volume by:

k

k^2

k^3

$3k$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Slant height	Apofema	Апофема
Lateral edge	Yon qirra	Боковое ребро
Frustum	Kesik jism	Усечённое тело
Sphere	Sfera	Сфера
Similar solids	O'xshash jismlar	Подобные тела
Concept map	Tushunchalar xaritasi	Карта понятий
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 needs the apothem because:

it is easier

lateral area uses the slant height

volume uses it

no reason

Q3's volume uses:

the slant l

the height h

the radius only

the sector angle

The frustum formula has:

two terms

three terms

one term

four terms

Volumes $27 : 125$ give heights in:

$27 : 125$

$3 : 5$

$9 : 25$

$1 : 5$

Quarter IV begins with:

new solids

applying these formulas in context

vectors

probability

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Square pyramid, base 10, height 12: the apothem

ANSWER 13

2. Same: lateral area

ANSWER 260

3. Same: total area

ANSWER 360

4. Same: volume

ANSWER 400

5. Cone $r = 5$, $l = 13$: height

ANSWER 12

6. Same cone: curved area

ANSWER 65π

7. Ball of area 144π : radius

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. Cone $r = 5$, $l = 13$: volume

ANSWER 100π

2. Same cone: the sector angle of its net

ANSWER $\approx 138.5^\circ$

3. Frustum $R = 6$, $r = 3$, $h = 4$: volume

ANSWER 84π

4. Same frustum: curved area

ANSWER 45π

5. Ball of area 144π : volume

ANSWER 288π

6. Two similar cones, volumes 27 and 125: the height ratio

ANSWER $3 : 5$

7. Same pair: the surface area ratio

ANSWER $9 : 25$

Hard · stretch and reason

7 PROBLEMS

1. Square pyramid, base 10, height 12: the angle of a lateral edge with the base

ANSWER $\approx 59.5^\circ$

2. Same pyramid: the dihedral angle along a base edge

ANSWER $\approx 67.4^\circ$

3. A cap of height 2 on a ball of radius 6: its volume

ANSWER $\frac{\pi}{3}(4)(18 - 2) = \frac{64\pi}{3} \approx 67.0$

4. A cone and a hemisphere on the same circle have equal volumes. Find $\frac{h}{r}$

ANSWER 2

5. A frustum has radii R and $\frac{R}{2}$ and height h . Find the fraction of the full cone it holds

ANSWER $\frac{7}{8}$

6. A sphere is inscribed in a cone of radius 6 and height 8. Find its radius

ANSWER $r = \frac{48}{16} = 3$

7. A ball is cut by two parallel planes 3 apart on a sphere of radius 10: the zone area

ANSWER 60π

07 Homework

Homework — 4 tasks

Bring the seven-formula page and the concept map to the first lesson of Quarter IV.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Write the seven formulas of Quarter III on one page in your own hand, each with a small sketch.

3. A cone of radius 9 cm and height 12 cm is cut halfway up by a plane parallel to the base. Find the volume and the curved area of the frustum that remains.
4. Write your target for Quarter IV in one checkable sentence, and date it.